Practice MCQs - 50 per week

Weeks 01 to 12 | VLSI Design Flow: RTL to GDS

600 newly authored multiple-choice questions: 50 for each course week. They are syllabus-based practice, not authenticated past exam questions or a prediction of the examination. They combine conceptual questions, applications, code traces and numerical checks.

Choose exactly one option per question. No answers are selected or printed. The numbered circles are fillable PDF controls. Save a copy after filling. For marks, use the matching browser practice file: finish an attempt, then choose Show marks. It displays the score without revealing correct answers.

In Boolean expressions, adjacent variable names denote AND, + denotes OR and a prime denotes complement unless a question states a different notation. Numerical assumptions and units belong to each question. Use the bookmarks or the week index below.

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Prepared for independent revision. Primary topic reference: official NPTEL course outline. Full source notes and provenance are in Resources/weekly.

Week 01 - IC foundations and pre-RTL design

50 practice MCQs | one point per question | no negative marking in this practice set

Week 01 | Question 01

Consider conventional CMOS ICs. A. Silicon is a semiconductor used in device regions. B. All interconnect layers must be electrically shorted together. C. Dielectrics separate conductors except at intended connections. Which statements are correct?

- 1. A only
- 2. A and C only
- 3. A, B and C
- 4. B and C only

Week 01 | Question 02

Two wires cross on insulated different layers with no via at the crossing. A. They need not be the same net. B. The crossing forces a DC short. C. Geometric separation proves zero capacitive coupling. Which statements are correct?

- 1. A and C only
- 2. A and B only
- 3. B and C only
- 4. A only

Week 01 | Question 03

Consider vias in an IC. A. They can connect selected metal layers. B. Their electrical and geometric properties matter in implementation. C. Adding a via automatically waives spacing rules. Which statements are correct?

- 1. A only
- 2. A and B only
- 3. B and C only
- 4. A, B and C

Week 01 | Question 04

What distinguishes a wafer from an individual die?

- 1. A die is the cylindrical silicon ingot before slicing.
- 2. A wafer usually contains many repeated die sites before dicing.
- 3. A wafer is the packaged form of a single die.
- 4. A die contains many separately packaged wafers.

Week 01 | Question 05

Under a simplified doubling-every-two-years transistor-count trend, what is the multiplication factor after six years?

- 1. 6
- 2. 8
- 3. 3
- 4. 12

Week 01 | Question 06

Which statement about a CMOS inverter is appropriate for an introductory static model?

- 1. The pull-up and pull-down networks drive opposite output logic levels.
- 2. Its metal layers replace the need for transistors.
- 3. Its output always equals its input.
- 4. Its input is connected directly to both supply rails.

Week 01 | Question 07

What does a photomask specify in photolithography?

- 1. The measured defect count after packaging.
- 2. The software instructions run by a processor.
- 3. The spatial pattern used for an exposure step.
- 4. The RTL execution schedule of the complete circuit.

Week 01 | Question 08

Which ordering is consistent with a simplified subtractive pattern-transfer sequence?

- 1. Develop resist, expose, deposit film, strip resist, coat resist.
- 2. Etch all layers, coat resist, package, expose, develop.
- 3. Deposit film, coat resist, expose, develop, etch, strip resist.
- 4. Expose bare silicon, etch, coat resist, deposit film, develop.

Week 01 | Question 09

Why is using one unchanged mask for all patterned IC layers generally incorrect?

- 1. Different layers require different geometric patterns and alignment.
- 2. Every layer must contain exactly the same conductor shape.
- 3. Masks describe only package outlines.
- 4. Metal layers do not require pattern definition.

Week 01 | Question 10

Which statement correctly separates deposition from lithography?

- 1. Deposition converts RTL into gates; lithography selects software.
- 2. Lithography always deposits the final interconnect metal.
- 3. Both terms refer only to testing packaged chips.
- 4. Deposition adds material; lithography defines a pattern in resist.

Week 01 | Question 11

Which responsibility best matches a fabless company?

- 1. Supply only packaging without doing design.
- 2. Design chips while contracting wafer fabrication to a foundry.
- 3. Manufacture masks but never specify a circuit.
- 4. Operate every fabrication step in its own wafer plant.

Week 01 | Question 12

What is the main purpose of a process design kit (PDK)?

- 1. Replace the need to define the product function.
- 2. Guarantee that any RTL description is physically manufacturable.
- 3. Provide process-related rules, models and design data.
- 4. Contain only the finished packaged chip inventory.

Week 01 | Question 13

A low-volume prototype must be field-updated often and shipped quickly. Which implementation is usually the first candidate?

- 1. An untested bare wafer because it is reprogrammable.
- 2. FPGA, subject to its power and performance limits.
- 3. A standard-cell ASIC because it has no fixed development cost.
- 4. A new full-custom ASIC regardless of its schedule.

Week 01 | Question 14

What does a standard-cell design methodology reuse?

- 1. Pre-designed and characterized cells assembled into a custom circuit.
- 2. One fixed complete chip for every application.
- 3. A fabrication line that never uses custom interconnect masks.
- 4. Only a general-purpose CPU software instruction set.

Week 01 | Question 15

Which design style generally offers the greatest transistor-level customization?

- 1. A standard off-the-shelf FPGA bitstream.
- 2. Software-only implementation on a purchased CPU.
- 3. A fixed gate array with no interconnect customization.
- 4. Full-custom design.

Week 01 | Question 16

Compared with an ASIC implementation of the same function, an FPGA often has which cost pattern?

- 1. No unit cost because it is programmable.
- 2. Higher application-specific fixed cost and zero unit cost.
- 3. Lower application-specific fixed cost but higher unit cost.
- 4. Identical fixed and variable costs at every volume.

Week 01 | Question 17

For total cost $C = F + N v$, what does the slope with respect to production quantity N represent?

- 1. Variable cost per unit v.
- 2. The number of masks in the process.
- 3. The fixed development cost F.
- 4. The fraction of dies passing test.

Week 01 | Question 18

An ASIC costs 1,200,000 + 20N and an FPGA solution costs 200,000 + 70N, in the same units. At what quantity do the costs match?

- 1. 50,000 units
- 2. 20,000 units
- 3. 25,000 units
- 4. 10,000 units

Week 01 | Question 19

Using C_ASIC=1,200,000+20N and C_FPGA=200,000+70N, which has lower modeled cost at N=10,000?

- 1. They both cost 1,400,000.
- 2. FPGA: 700,000 versus ASIC: 1,200,000.
- 3. ASIC: 400,000 versus FPGA: 900,000.
- 4. FPGA: 900,000 versus ASIC: 1,400,000.

Week 01 | Question 20

A wafer costs 6,000 units, has 500 gross dies and 80% die yield. Ignore all other costs. What is fabrication cost per good die?

- 1. 15 units
- 2. 10 units
- 3. 12 units
- 4. 24 units

Week 01 | Question 21

An ASIC requires more fixed investment and has unit cost equal to its FPGA competitor. What happens in the simple linear cost model?

- 1. The ASIC becomes cheaper after exactly one unit.
- 2. There is no positive-volume break-even from unit-cost savings.
- 3. The two total costs are identical at every volume.
- 4. The larger fixed investment vanishes at high volume.

Week 01 | Question 22

If 350 out of 500 fabricated dies pass the stated tests, what is die yield?

- 1. 70%
- 2. 143%
- 3. 30%
- 4. 0.7%

Week 01 | Question 23

Which comparison distinguishes power from energy?

- 1. A lower-power implementation always uses less task energy.
- 2. Energy per task depends on both average power and task duration.
- 3. Clock frequency alone determines both power and energy.
- 4. Power and energy have the same physical units.

Week 01 | Question 24

For unchanged activity, capacitance and frequency, voltage falls from 1.0 V to 0.8 V. In P_switch proportional to V squared, the power ratio is what?

- 1. 0.64
- 2. 0.80
- 3. 0.36
- 4. 1.25

Week 01 | Question 25

Which design conclusion is justified by an increased clock frequency alone?

- 1. It always reduces the number of cycles per task.
- 2. It guarantees lower energy per completed task.
- 3. It proves smaller die area.
- 4. It reduces clock period, but end-to-end task latency also depends on cycle count.

Week 01 | Question 26

As a design moves from an algorithmic model toward physical layout, what generally happens?

- 1. Abstraction increases while geometry disappears.
- 2. All models contain exactly the same detail.
- 3. Exploration necessarily becomes faster at every later stage.
- 4. Abstraction decreases and implementation detail increases.

Week 01 | Question 27

Why can higher-level models help early design exploration?

- 1. They prove every physical signoff condition automatically.
- 2. They remove all need for functional specifications.
- 3. They omit lower-level detail so alternative architectures can be evaluated sooner.
- 4. They require completed mask data before simulation.

Week 01 | Question 28

Which function is usually a suitable candidate to remain in software?

- 1. A proven compute bottleneck needing fixed high-throughput arithmetic in every case.
- 2. The photoresist exposure step.
- 3. A control policy that must be updated frequently.
- 4. The mandatory physical connection between two metal layers.

Week 01 | Question 29

A computation takes 100 ms in software. Dedicated hardware needs 10 ms plus 30 ms transfer overhead. What is the end-to-end speedup?

- 1. 3.33 times
- 2. 0.4 times
- 3. 2.5 times
- 4. 10 times

Week 01 | Question 30

Before accelerating a computation, why should transfer and coordination overhead be included?

- 1. They affect only the package color.
- 2. They are unrelated to latency.
- 3. They can reduce or eliminate the expected end-to-end speedup.
- 4. They are always zero when using dedicated hardware.

Week 01 | Question 31

What makes a system-on-chip (SoC) more than a collection of isolated verified IP blocks?

- 1. The integrated system also needs compatible interfaces and system-level verification.
- 2. Every block must be rewritten from scratch.
- 3. Top-level verification becomes unnecessary after vendor verification.
- 4. No block may be reused.

Week 01 | Question 32

Which statement distinguishes soft IP from hard IP in a common usage?

- 1. Soft IP is typically supplied as synthesizable RTL; hard IP as a process-specific physical implementation.
- 2. Soft IP contains no functional behavior.
- 3. Hard IP can never contain a memory.
- 4. Both necessarily have identical process portability.

Week 01 | Question 33

An IP block has a different reset convention and clock domain from its new host. What is required?

- 1. Assume the vendor verified every possible host environment.
- 2. Remove all registers from the IP.
- 3. Resolve the integration assumptions and verify the interface behavior.
- 4. Connect the ports by name and omit further checks.

Week 01 | Question 34

Which item is useful IP integration metadata?

- 1. Only a screenshot of the completed chip package.
- 2. Interface timing, protocol, clock/reset and configuration requirements.
- 3. Only the number of staff employed by the vendor.
- 4. Only the marketing name of the block.

Week 01 | Question 35

What is behavioural synthesis, also called high-level synthesis (HLS)?

- 1. Constructing a timed RTL architecture from an algorithmic description under constraints.
- 2. Executing a C program only on a general-purpose CPU.
- 3. Drawing final transistor polygons directly from an SDC file.
- 4. Converting a packaged chip back into a silicon ingot.

Week 01 | Question 36

Which decision belongs to HLS scheduling?

- 1. Assign operations to clock steps while respecting dependencies and constraints.
- 2. Measure wafer yield after manufacturing.
- 3. Choose the final chip package color.
- 4. Assign every net to a detailed routing track.

Week 01 | Question 37

Which decision belongs to HLS resource binding?

- 1. Choose which lithography mask is exposed first.
- 2. Compute final layout-vs-schematic equivalence from polygons.
- 3. Specify only the total production volume.
- 4. Associate scheduled operations with particular hardware resources.

Week 01 | Question 38

In HLS terminology, what does resource allocation determine?

- 1. The physical routing layer of every signal.
- 2. Only which die passes a manufacturing test.
- 3. Only the final names of input variables.
- 4. The types and numbers of resources made available to the architecture.

Week 01 | Question 39

A computation needs two additions. One nonpipelined adder can perform one addition per cycle. Ignoring other limits, what resource lower bound applies to its steady-state initiation interval?

- 1. At least 2 cycles per transaction.
- 2. At most half a cycle.
- 3. Exactly 1 cycle because the expression has one result.
- 4. Zero cycles if the RTL is automatically generated.

Week 01 | Question 40

A two-stage pipeline runs at a 10 ns period and accepts a new independent item every cycle. What are its ideal latency and initiation interval in time?

- 1. 10 ns latency and 20 ns initiation interval.
- 2. 10 ns latency and zero initiation interval.
- 3. 20 ns latency and 10 ns initiation interval.
- 4. 20 ns latency and 20 ns initiation interval necessarily.

Week 01 | Question 41

A pipeline has a 5 ns clock and an initiation interval of 2 cycles. What is its ideal steady-state throughput?

- 1. 200 million items per second
- 2. 10 million items per second
- 3. 100 million items per second
- 4. 400 million items per second

Week 01 | Question 42

What can resource sharing in HLS change?

- 1. Area, operand selection, storage/control needs and the acceptance schedule.
- 2. The need for any clocked storage always disappears.
- 3. Only variable spelling, with no architectural effect.
- 4. Only fabrication yield, never timing.

Week 01 | Question 43

Which is an advantage of HLS when used with suitable code and constraints?

- 1. It can explore different RTL architectures from one high-level function.
- 2. It accepts every operating-system call as hardware.
- 3. It eliminates all RTL and system verification.
- 4. It guarantees the best possible PPA for every algorithm.

Week 01 | Question 44

Which limitation should be considered before using an HLS tool?

- 1. Only a supported synthesizable language subset maps to finite hardware resources.
- 2. All loop bounds and data dependencies are irrelevant.
- 3. Any arbitrary host operating-system operation becomes a hardware port automatically.
- 4. The generated circuit can dynamically acquire unlimited physical memory.

Week 01 | Question 45

A pipelined implementation computes (a+b)+c from successive input transactions. Why may c need a matching delay register?

- 1. To force every transaction to have identical input values.
- 2. To eliminate the second addition mathematically.
- 3. To convert the entire design into software.
- 4. To align c with the partial sum from the same transaction.

Week 01 | Question 46

A shell is currently in /home/k/lab. What path does ../rtl/top.v refer to?

- 1. /rtl/top.v
- 2. /home/rtl/top.v
- 3. /home/k/lab/rtl/top.v
- 4. /home/k/rtl/top.v

Week 01 | Question 47

Consider touch under normal Unix permissions. A. It can create a file if absent. B. It normally erases an existing file. C. It can update timestamps while preserving existing contents. Which statements are correct?

- 1. A and C only
- 2. A and B only
- 3. A, B and C
- 4. B only

Week 01 | Question 48

Consider common Unix commands. A. ls -a includes dot-prefixed entries. B. df primarily lists shell job numbers. C. du can report directory disk usage. Which statements are correct?

- 1. B and C only
- 2. A, B and C
- 3. A only
- 4. A and C only

Week 01 | Question 49

What distinguishes Ctrl+Z followed by bg from terminating a foreground job?

- 1. The operating system is put to sleep.
- 2. The job is suspended and then resumed in the background.
- 3. The process is permanently deleted from disk.
- 4. The command is rerun from its beginning with a new input file.

Week 01 | Question 50

Which pairing is correct for common Unix tools?

- 1. df lists shell job numbers; du lists register names.
- 2. du reports clock period; df reports gate delay.
- 3. du reports directory disk usage; df reports filesystem space.
- 4. du and df both change file ownership.

Week 02 - Implementation, verification, manufacturing and Tcl

50 practice MCQs | one point per question | no negative marking in this practice set

Week 02 | Question 01

Consider ASIC logic synthesis. A. RTL is a usual input. B. A target cell library provides implementation choices. C. The normal output is a fabricated packaged chip. Which statements are correct?

- 1. A, B and C
- 2. A and C only
- 3. A and B only
- 4. B only

Week 02 | Question 02

Consider constrained synthesis inputs. A. Clock targets can influence implementation. B. Cell timing models can influence mapping. C. A testbench waveform alone defines all target-library behavior. Which statements are correct?

- 1. A and B only
- 2. A only
- 3. B and C only
- 4. C only

Week 02 | Question 03

Consider technology mapping. A. It selects available cell implementations. B. Equivalent logic may have different PPA in different libraries. C. It determines every final routed wire polygon. Which statements are correct?

- 1. B and C only
- 2. A and B only
- 3. A and C only
- 4. A, B and C

Week 02 | Question 04

A library cell has input pins A and B and output Y. Three instances are used. How many instance input pins are present?

- 1. 2
- 2. 6
- 3. 9
- 4. 3

Week 02 | Question 05

What distinguishes a cell master from an instance?

- 1. A master defines a reusable cell; an instance is one occurrence in the design.
- 2. A master is always a complete chip package.
- 3. Each instance must have a different Boolean function from its master.
- 4. An instance cannot have pin connections.

Week 02 | Question 06

Why may synthesis choose two drive strengths for cells implementing the same logic function?

- 1. To avoid using any technology library.
- 2. To remove all input capacitance from the circuit.
- 3. To change the truth table of the function.
- 4. To meet different electrical loads and timing requirements.

Week 02 | Question 07

Which item primarily records logical connections among instances?

- 1. A packaging invoice.
- 2. A netlist.
- 3. A photomask inventory.
- 4. A wafer map showing failed dies.

Week 02 | Question 08

Which statement separates logical and physical representations?

- 1. A netlist gives connectivity; layout also gives geometry and layer information.
- 2. They are identical descriptions in every respect.
- 3. A netlist always contains final detailed wire shapes.
- 4. A layout never represents connectivity.

Week 02 | Question 09

Which stage is normally performed before standard-cell placement?

- 1. Post-silicon binning.
- 2. Final package shipment.
- 3. Wafer dicing.
- 4. Floorplanning or chip planning.

Week 02 | Question 10

What does placement primarily determine?

- 1. Which production dies pass wafer test.
- 2. Locations of cells within the physical design.
- 3. The exact transistor doping profile for fabrication.
- 4. The source-code grammar of Verilog.

Week 02 | Question 11

What is a primary purpose of clock-tree synthesis?

- 1. Convert all signal nets into power rails.
- 2. Distribute clock signals to sinks with controlled timing and electrical behavior.
- 3. Remove every sequential element.
- 4. Prove all functional properties without a specification.

Week 02 | Question 12

What does routing add to a placed netlist?

- 1. A different required product specification.
- 2. Wafer yield measurements.
- 3. A new high-level software algorithm only.
- 4. Physical wire and via connections.

Week 02 | Question 13

Which flow ordering is normally appropriate?

- 1. Route, dice wafer, floorplan, place.
- 2. Package, route, synthesize, specify behavior.
- 3. Floorplan, place, build clock network, route.
- 4. Clock-tree synthesis, fabrication, RTL coding, floorplanning.

Week 02 | Question 14

What does DRC check?

- 1. The final market price of the device.
- 2. Geometric compliance with process design rules.
- 3. Whether every simulation stimulus was generated randomly.
- 4. Equality of all software execution times.

Week 02 | Question 15

What does LVS primarily compare?

- 1. The source-code indentation of two modules.
- 2. Connectivity extracted from layout with the reference circuit connectivity.
- 3. The purchase prices of two FPGA boards.
- 4. Only the lengths of file names.

Week 02 | Question 16

What is tapeout in the design-to-manufacture flow?

- 1. Packaging a die after final test.
- 2. Running a behavioral simulation for the first time.
- 3. Release of the finalized design data for fabrication.
- 4. Powering off all EDA workstations.

Week 02 | Question 17

What distinguishes functional verification from manufacturing test?

- 1. Manufacturing test proves that the specification itself is correct.
- 2. Functional verification is performed only after wafer fabrication.
- 3. Verification checks the design against intended behavior; manufacturing test targets fabricated-device defects.
- 4. Both exclusively measure package weight.

Week 02 | Question 18

Which conclusion follows from passing a finite set of simulation tests?

- 1. The tested behaviors passed; untested behaviors may still contain errors.
- 2. All physical design rules have necessarily passed.
- 3. The chip is guaranteed to have 100% manufacturing yield.
- 4. Every possible input and state has necessarily been proved.

Week 02 | Question 19

What is a testbench used for?

- 1. Applying stimuli and checking or observing the design responses.
- 2. Fabricating silicon transistors.
- 3. Defining every detailed routing track.
- 4. Measuring the market demand for the chip.

Week 02 | Question 20

What does static timing analysis primarily check?

- 1. All functional properties of arbitrary software.
- 2. Timing requirements over modeled timing paths without enumerating functional stimulus sequences.
- 3. The physical strength of the chip package.
- 4. Only the color coding in waveform viewers.

Week 02 | Question 21

Which quantity is a timing constraint rather than a simulation stimulus?

- 1. The expected text printed by a testbench.
- 2. A randomly selected packet payload.
- 3. One input vector applied at 12 ns.
- 4. Required clock period.

Week 02 | Question 22

Which statement about setup and hold is appropriate?

- 1. Setup applies only to combinational circuits without clocks.
- 2. Both are solely restrictions on package soldering.
- 3. Hold is always solved by increasing clock period.
- 4. Setup constrains data before capture; hold constrains data after capture.

Week 02 | Question 23

What does positive setup slack indicate for the modeled check?

- 1. Hold timing necessarily fails.
- 2. All functional verification is complete.
- 3. The required clock is necessarily the fastest possible physical clock.
- 4. Arrival meets the required setup boundary with margin.

Week 02 | Question 24

If 720 of 900 fabricated dies pass the stated test, what is the yield?

1. 125%
2. 72%
3. 80%
4. 20%

Week 02 | Question 25

Using Y=exp(-D A), with D A = 0, what is the modeled defect-limited yield?

1. 0
2. It is undefined.
3. 0.5
4. 1

Week 02 | Question 26

Under Y=exp(-D A), increasing area A while keeping positive D fixed has what effect?

1. Decreases yield in this model.
2. Leaves yield unchanged for all A.
3. Changes yield only after packaging.
4. Increases yield in this model.

Week 02 | Question 27

For Y=exp(-D A), D=0.2 defects per square centimetre and A=1 square centimetre, yield is approximately what?

1. 98.0%
2. 18.1%
3. 81.9%
4. 20.0%

Week 02 | Question 28

If D is expressed per square centimetre, which area value should be paired with it for a 100 square millimetre die?

1. 0.01 square centimetre.
2. 100 square centimetres.
3. 10 square centimetres.
4. 1 square centimetre.

Week 02 | Question 29

A 10,000-unit wafer has 400 gross dies and 50% yield. Ignore other costs. Cost per good die is what?

1. 25 units
2. 50 units
3. 100 units
4. 20 units

Week 02 | Question 30

What does a resolution enhancement technique primarily address in lithography?

1. Faithful printing of small intended geometric features.
2. The clock period in an RTL simulation.
3. The storage capacity of a logic analyzer.
4. The number of Verilog module parameters.

Week 02 | Question 31

Which item is associated with optical lithography resolution enhancement?

- 1. RTL constant propagation.
- 2. Optical proximity correction.
- 3. Clock-domain crossing synchronization.
- 4. Scan-chain shifting.

Week 02 | Question 32

Which is a correct distinction between synthesis and physical design?

- 1. Synthesis directly packages every manufactured die.
- 2. Synthesis builds a logical implementation; physical design realizes it geometrically.
- 3. Physical design must be completed before any RTL can be written.
- 4. The two stages always take exactly the same inputs and produce the same files.

Week 02 | Question 33

Why can physical design require iteration after a logically correct netlist exists?

- 1. Logical correctness forces every cell to occupy the same location.
- 2. Routing cannot change delay.
- 3. Wire parasitics, congestion and physical rules may prevent implementation targets from being met.
- 4. Layout contains no process-dependent constraints.

Week 02 | Question 34

What is parasitic extraction used for?

- 1. Assigning software licenses to students.
- 2. Choosing source-code keywords.
- 3. Measuring only the number of staff in a design team.
- 4. Deriving interconnect electrical models from physical geometry.

Week 02 | Question 35

What does a PDK describe more directly than an RTL source file?

- 1. The exact cycle-by-cycle behavior of every customer design.
- 2. The application software user interface.
- 3. The pass/fail history of every future die.
- 4. The target manufacturing process and associated design information.

Week 02 | Question 36

In Tcl, what does set x 7 do?

- 1. Assign the value 7 to variable x.
- 2. Define a seven-bit Verilog wire.
- 3. Print the current directory seven times.
- 4. Create seven operating-system processes.

Week 02 | Question 37

In Tcl, what value does set y $x use when x has the value 7?

- 1. Always an empty string.
- 2. The literal characters dollar-sign x.
- 3. The current value of x.
- 4. The name of the last file opened.

Week 02 | Question 38

In Tcl, what is the main effect of braces in set s {$x}?

- 1. They execute x as a shell command.
- 2. They suppress variable substitution inside this word.
- 3. They convert x to a Verilog parameter.
- 4. They delete x after assigning s.

Week 02 | Question 39

In Tcl, square brackets around a command normally request what?

- 1. A mandatory array slice.
- 2. A physical routing obstruction.
- 3. A comment block.
- 4. Command substitution.

Week 02 | Question 40

What is the result of expr {3 + 4 * 2} in Tcl?

- 1. 11
- 2. 24
- 3. 14
- 4. 10

Week 02 | Question 41

In Tcl, which command writes text to standard output?

- 1. display_netlist
- 2. echo_signal
- 3. append-only
- 4. puts

Week 02 | Question 42

For set xs {4 7 9}, what does lindex $xs 1 return?

- 1. 9
- 2. 4
- 3. 1
- 4. 7

Week 02 | Question 43

For set xs {4 7 9}, what does llength $xs return?

- 1. 3
- 2. 2
- 3. 9
- 4. 20

Week 02 | Question 44

Which Tcl construct iterates over elements of a list?

- 1. foreach
- 2. wire
- 3. always_ff
- 4. module

Week 02 | Question 45

In Tcl, which command appends elements to a list variable?

- 1. initial
- 2. assign
- 3. lappend
- 4. link_library

Week 02 | Question 46

What does Tcl proc define?

- 1. A named procedure.
- 2. A Unix directory permission.
- 3. A physical clock sink.
- 4. A standard-cell timing arc.

Week 02 | Question 47

Consider a Tcl procedure. A. proc defines it. B. return can provide its result. C. puts and return always have identical effects on the caller. Which statements are correct?

- 1. B only
- 2. A and B only
- 3. A, B and C
- 4. A and C only

Week 02 | Question 48

Consider Tcl in EDA tools. A. A tool may add domain-specific commands. B. Every base Tcl interpreter contains every EDA command. C. A command can change the active design database. Which statements are correct?

- 1. A and C only
- 2. B and C only
- 3. A, B and C
- 4. A only

Week 02 | Question 49

If a Tcl list has 6 elements and a loop processes each element once, how many body executions occur?

- 1. 7
- 2. 12
- 3. 6
- 4. 5

Week 02 | Question 50

Why should an EDA script check that required input files exist?

- 1. To force all cells to have one drive strength.
- 2. To distinguish missing or mislocated inputs from later tool failures.
- 3. To guarantee zero delay after routing.
- 4. To eliminate all verification requirements.

Week 03 - Verilog, simulation and high-level synthesis

50 practice MCQs | one point per question | no negative marking in this practice set

Week 03 | Question 01

Consider Verilog modules. A. One definition can have several instances. B. Parameters can configure instances at elaboration. C. Every instance must use a different source language. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. A and C only
- 4. B only

Week 03 | Question 02

Consider elaboration. A. It resolves the design hierarchy. B. It resolves parameter-dependent structure. C. Successful elaboration proves all functional requirements. Which statements are correct?

- 1. C only
- 2. B and C only
- 3. A only
- 4. A and B only

Week 03 | Question 03

Consider traditional Verilog data objects. A. A reg declaration always guarantees a flip-flop. B. A wire represents a driven net. C. A procedural reg can model a combinational result. Which statements are correct?

- 1. A and C only
- 2. B and C only
- 3. A and B only
- 4. A, B and C

Week 03 | Question 04

Which values are in Verilog four-state logic?

- 1. low, high, rise and fall only.
- 2. 0, 1, 2 and 3 only.
- 3. 0, 1, x and z.
- 4. false and true only.

Week 03 | Question 05

What does z normally represent on a Verilog net?

- 1. High impedance.
- 2. A guaranteed logic zero.
- 3. A decimal integer of value 26.
- 4. A guaranteed logic one.

Week 03 | Question 06

What does x generally indicate in a Verilog simulation?

- 1. A request to stop simulation.
- 2. A bit that must be ignored by all operators.
- 3. An unknown logic value.
- 4. Always a legal physical third supply voltage.

Week 03 | Question 07

What is the value of the reduction OR expression |4'b0000?

1. 1'b1
2. 1'b0
3. 4'b0000
4. 1'bz

Week 03 | Question 08

What is the value of the reduction AND expression &4'b1110?

1. 4'b1110
2. 1'b1
3. 1'b0
4. 1'bz

Week 03 | Question 09

What is 4'b1010 & 4'b1100?

1. 4'b0000
2. 4'b1110
3. 4'b1000
4. 4'b0110

Week 03 | Question 10

What is 4'b1010 | 4'b1100?

1. 4'b1110
2. 4'b0010
3. 4'b0110
4. 4'b1000

Week 03 | Question 11

What is 4'b1010 ^ 4'b1100?

1. 4'b0000
2. 4'b0110
3. 4'b1000
4. 4'b1110

Week 03 | Question 12

What is the value of logical AND for 4'b0010 && 4'b0100?

1. 1'b1
2. 4'b0000
3. 1'b0
4. 4'b0110

Week 03 | Question 13

How many bits are in {2'b10,2'b01}?

1. 4
2. 8
3. 2
4. 1

Week 03 | Question 14

What is {2{2'b10}}?

1. 2'b10
2. 4'b1010
3. 4'b1111
4. 4'b0101

Week 03 | Question 15

For reg [7:0] v = 8'b11001010, what is v[3:0]?

1. 4'b0010
2. 4'b1010
3. 4'b1100
4. 4'b0101

Week 03 | Question 16

What unsigned decimal value does 8'h2F denote?

1. 47
2. 31
3. 32
4. 63

Week 03 | Question 17

What is the stored 4-bit result of 4'b1111 + 4'b0001 when assigned to a 4-bit destination?

1. 4'b1111
2. 5'b10000
3. 4'b0000
4. 4'b0001

Week 03 | Question 18

What is the signed two's-complement value of 4'b1110 when explicitly treated as signed?

1. -1
2. -2
3. 2
4. 14

Week 03 | Question 19

Which construct continuously drives a net from an expression?

1. parameter only
2. assign
3. task only
4. initial only

Week 03 | Question 20

Which statement about an initial block is correct?

1. It starts once at simulation time zero for that instance.
2. It must repeat on every clock edge.
3. It always infers a clocked register.
4. It can contain only one assignment.

Week 03 | Question 21

Which behavior is described by always @(posedge clk)?

- 1. Execute once before time zero.
- 2. Execute only when clk stays low for one cycle.
- 3. Execute the block when a positive edge event occurs on clk.
- 4. Execute at every integer simulation time regardless of clk.

Week 03 | Question 22

Within one procedural sequence, what distinguishes blocking assignment =?

- 1. It always waits until the next clock cycle.
- 2. It is legal only in continuous assignments.
- 3. It never changes a variable value.
- 4. It updates before the following statement in that sequence proceeds.

Week 03 | Question 23

What does a nonblocking assignment normally schedule?

- 1. Immediate execution of a new operating-system thread.
- 2. An update for a later simulation scheduling region of the current time step.
- 3. A permanent prohibition on reading the destination.
- 4. A mandatory delay of one full clock period.

Week 03 | Question 24

Before a clock edge a=0 and b=1. The block executes a <= b; b <= a;. What are the updated values after nonblocking updates?

- 1. a=0, b=0
- 2. a=1, b=1
- 3. a=1, b=0
- 4. a=0, b=1

Week 03 | Question 25

Before a clock edge a=0 and b=1. In one block a = b; b = a; executes in that order. What are the final values?

- 1. a=0, b=1
- 2. a=1, b=0
- 3. a=0, b=0
- 4. a=1, b=1

Week 03 | Question 26

Why can two clocked blocks that communicate through blocking assignments race?

- 1. Every blocking assignment waits one cycle.
- 2. Verilog always sorts blocks by signal name.
- 3. A race requires a fabricated chip.
- 4. Their active-region execution order may affect the observed result.

Week 03 | Question 27

What does $display normally do?

- 1. Automatically wait until all nonblocking updates complete.
- 2. Set every unknown value to zero.
- 3. Print values when the statement executes.
- 4. Synthesize an output display controller.

Week 03 | Question 28

What is a useful distinction for $strobe?

- 1. It reports values at the end of the current time step.
- 2. It changes register values to their reset state.
- 3. It is the only way to define a clock.
- 4. It runs only after the simulation process exits.

Week 03 | Question 29

With `timescale 1ns/1ps, a delay #2.5 represents what nominal time?

- 1. 250 ns
- 2. 2.5 microseconds
- 3. 2.5 ps
- 4. 2.5 ns

Week 03 | Question 30

A clock generator toggles every #5 under a 1 ns time unit. What is its period?

- 1. 10 ns
- 2. 5 ns
- 3. 1 ns
- 4. 20 ns

Week 03 | Question 31

For a 10 ns ideal clock period, what is the frequency?

- 1. 200 MHz
- 2. 1 GHz
- 3. 100 MHz
- 4. 10 MHz

Week 03 | Question 32

Why initialize a simulated clock before repeatedly complementing it?

- 1. Initialization makes the clock asynchronous automatically.
- 2. An initialized clock cannot toggle.
- 3. It changes all timescale declarations in other modules.
- 4. Complementing an unknown value can leave it unknown.

Week 03 | Question 33

Which task commonly ends a Verilog simulation?

- 1. $finish
- 2. $time
- 3. $monitor
- 4. $dumpvars

Week 03 | Question 34

What is the purpose of a VCD file?

- 1. Replace an RTL testbench with a technology library.
- 2. Define a wafer defect distribution.
- 3. Store final detailed routing geometry.
- 4. Record selected signal value changes for later waveform inspection.

Week 03 | Question 35

What does statement coverage measure?

- 1. Which executable statements were exercised.
- 2. Whether every possible input sequence was proved.
- 3. Whether the specification contains no omissions.
- 4. Whether every physical net is routed legally.

Week 03 | Question 36

What is a limitation of 100% code coverage?

- 1. It proves that no assertion is needed.
- 2. It proves that every manufactured chip works.
- 3. It guarantees minimum power.
- 4. It does not by itself prove functional correctness or complete specification coverage.

Week 03 | Question 37

Why use a self-checking testbench?

- 1. It forces every design to be combinational.
- 2. It compares observed results with expected behavior automatically.
- 3. It removes the need to specify expected behavior.
- 4. It fabricates test structures.

Week 03 | Question 38

What is the role of a reference model in a testbench?

- 1. Generate final masks.
- 2. Supply expected results against which the DUT is checked.
- 3. Replace all input stimuli.
- 4. Define only the chip outline.

Week 03 | Question 39

Which statement separates compilation from simulation success?

1. A design may compile while still containing functional errors.
2. Every syntax error is a manufacturing defect.
3. Compilation proves every input-output relation.
4. Simulation can run only on a physically fabricated design.

Week 03 | Question 40

What is the result of 1'bx === 1'bx in Verilog?

1. 1'b1
2. 1'b0
3. 1'bx
4. 1'bz

Week 03 | Question 41

What is the result of 1'bx == 1'bx in ordinary four-state logical equality?

1. 1'b1
2. 1'bz
3. 1'bx
4. 1'b0

Week 03 | Question 42

With sel=1'bx, what is the bitwise-merged value of sel ? 4'b1010 : 4'b1000?

1. 4'b10x0
2. 4'bxxxx
3. 4'b1000
4. 4'b1010

Week 03 | Question 43

What is a Verilog parameter commonly used for?

1. A physical via coordinate.
2. A runtime file descriptor only.
3. An elaboration-time configurable constant.
4. A mandatory register updated each cycle.

Week 03 | Question 44

What is a generate construct primarily used for?

1. Recording package test results.
2. Repeating a software computation once per simulation second.
3. Defining optical exposure conditions.
4. Conditionally or repetitively constructing hardware at elaboration.

Week 03 | Question 45

For a task and a function in traditional synthesizable Verilog usage, which statement is appropriate?

- 1. A function always creates a new clock domain.
- 2. A function must contain a # delay.
- 3. A task cannot have output arguments.
- 4. A function is used as an expression and should not consume simulation time.

Week 03 | Question 46

What does high-level synthesis with Bambu aim to generate from a supported high-level description?

- 1. A text editor configuration.
- 2. A hardware RTL implementation.
- 3. A silicon wafer with no downstream tools.
- 4. Only a waveform screenshot.

Week 03 | Question 47

Compare an HLS algorithm and its generated RTL. A. They can have different timing descriptions. B. Transaction acceptance must be interpreted consistently. C. Matching a few constant-input results proves all streaming alignments. Which statements are correct?

- 1. B and C only
- 2. A and B only
- 3. A, B and C
- 4. A only

Week 03 | Question 48

Consider cycle latency. A. It needs a defined input-acceptance event. B. It needs a corresponding output-valid event. C. It is necessarily equal to the initiation interval. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. A and C only
- 4. B only

Week 03 | Question 49

If an RTL pipeline produces valid outputs every 3 cycles at 100 MHz, what is its steady-state throughput?

- 1. 300 million outputs per second.
- 2. About 3.33 million outputs per second.
- 3. 100 million outputs per second.
- 4. About 33.3 million outputs per second.

Week 03 | Question 50

Which statement about simulation time and wall-clock time is correct?

- 1. Simulated nanoseconds need not correspond to the same elapsed real time.
- 2. Faster simulation changes the specified RTL delays.
- 3. Every simulated nanosecond must take one real nanosecond.
- 4. Wall-clock speed proves the circuit frequency.

Week 04 - RTL synthesis and two-level logic optimization

50 practice MCQs | one point per question | no negative marking in this practice set

Week 04 | Question 01

Consider incomplete assignments in combinational RTL. A. Retaining an old output value may imply storage. B. A missing else always implements a clock-edge flip-flop. C. Assigning all outputs on all paths helps avoid unintended latches. Which statements are correct?

- 1. A and C only
- 2. A and B only
- 3. B only
- 4. A, B and C

Week 04 | Question 02

Consider combinational modeling. A. Default assignments can cover otherwise unassigned branches. B. An incomplete sensitivity list can cause stale simulation behavior. C. Merely declaring an output reg guarantees a register is intended. Which statements are correct?

- 1. A, B and C
- 2. A and B only
- 3. B and C only
- 4. A only

Week 04 | Question 03

Consider an edge-triggered register. A. Its state updates at its specified active clock event. B. Its RTL must be modeled with timing intent. C. An unconditional combinational assign alone necessarily creates that state. Which statements are correct?

- 1. C only
- 2. A and B only
- 3. B and C only
- 4. A and C only

Week 04 | Question 04

What does if (sel) y=a; else y=b; describe in a complete combinational block?

- 1. An oscillator regardless of inputs.
- 2. A fabrication mask selector.
- 3. A multiplexer function.
- 4. A mandatory four-stage pipeline.

Week 04 | Question 05

For a 4-bit unsigned input x, what function can y=x*4 implement when y is wide enough?

- 1. A compulsory general-purpose CPU.
- 2. A divide-by-four network.
- 3. A right shift by four bits.
- 4. A two-bit left shift with appropriate zero extension.

Week 04 | Question 06

Why are signal widths important in synthesis?

- 1. Wider signals always use less area.
- 2. They affect represented values, truncation and the amount of arithmetic hardware.
- 3. They affect only source-code file size.
- 4. They never affect overflow behavior.

Week 04 | Question 07

What happens to a constant false branch during ordinary optimization?

- 1. It must occupy the same area as the whole design.
- 2. It can be removed if it has no required observable effect.
- 3. It becomes a new input clock.
- 4. It is converted into a package pin.

Week 04 | Question 08

Why can a statically bounded for loop in synthesizable RTL describe parallel hardware?

- 1. Every loop iteration must execute on an embedded CPU.
- 2. The source keyword for directly sets clock frequency.
- 3. RTL loops always consume one clock cycle per iteration.
- 4. Elaboration or synthesis can expand the iterations into structural operations.

Week 04 | Question 09

What is the output of synthesizing a combinational constant assignment y=1'b0, subject to normal technology implementation?

- 1. An arbitrary unknown output.
- 2. A programmable oscillator.
- 3. A clocked counter.
- 4. A constant-zero connection, potentially using a tie cell.

Week 04 | Question 10

Which transformation is constant propagation?

- 1. Replacing expressions whose operands are known constants with their computed values.
- 2. Extracting metal capacitance.
- 3. Moving a standard cell to a new coordinate.
- 4. Adding clock buffers at every input.

Week 04 | Question 11

In Boolean algebra, what is x OR 0?

- 1. 1
- 2. NOT x
- 3. x
- 4. 0

Week 04 | Question 12

In Boolean algebra, what is x AND 0?

1. x
2. 0
3. NOT x
4. 1

Week 04 | Question 13

What is x OR (x AND y)?

1. x XOR y
2. x
3. y
4. NOT x

Week 04 | Question 14

What is NOT(x AND y), by De Morgan's law?

1. (NOT x) AND (NOT y).
2. x XOR y.
3. (NOT x) OR (NOT y).
4. x OR y.

Week 04 | Question 15

For three inputs x,y,z, F is 1 at minterm numbers 1,3,5,7, where x is the most significant bit. What is F?

1. z
2. x
3. NOT z
4. y

Week 04 | Question 16

For three inputs x,y,z, F is 1 at minterms 2,3,6,7. What is F?

1. y
2. NOT y
3. x
4. z

Week 04 | Question 17

A two-input XOR is 1 on which input combinations?

1. 00 and 01.
2. 01 and 10.
3. 10 and 11.
4. 00 and 11.

Week 04 | Question 18

What is an implicant of a Boolean function in a sum-of-products setting?

1. A term that must cover every OFF-set point.
2. Only a minterm containing every variable complemented.
3. Any product term with the largest number of variables.
4. A product term whose covered assignments do not include an OFF-set point.

Week 04 | Question 19

What makes an implicant prime?

- 1. It cannot be expanded further by removing a literal without covering an OFF-set point.
- 2. It contains a prime number of variables.
- 3. It is always the only valid implementation.
- 4. It covers exactly one minterm in every function.

Week 04 | Question 20

What makes a prime implicant essential?

- 1. It has the longest source-code expression.
- 2. It appears first alphabetically.
- 3. It contains every variable in the function.
- 4. It covers at least one ON-set minterm covered by no other prime implicant.

Week 04 | Question 21

What is the role of a don't-care condition in minimization?

- 1. It permits wrong behavior on all specified ON-set points.
- 2. Its output may be chosen to simplify implementation within the stated specification.
- 3. It automatically means the input can never occur physically.
- 4. It must always be implemented as an unknown voltage.

Week 04 | Question 22

What must a valid SOP cover include?

- 1. Only the don't-care points.
- 2. Every required ON-set assignment while avoiding required OFF-set assignments.
- 3. Every OFF-set point.
- 4. Exactly one term regardless of the function.

Week 04 | Question 23

What is a literal?

- 1. Only a decimal constant larger than one.
- 2. A physical cell instance.
- 3. A variable or its complement appearing in an expression.
- 4. A simulation waveform marker.

Week 04 | Question 24

How many literal occurrences appear in ab + ac?

- 1. 2
- 2. 4
- 3. 3
- 4. 6

Week 04 | Question 25

How many minterms are in a complete truth table for 4 Boolean inputs?

- 1. 8
- 2. 16
- 3. 32
- 4. 4

Week 04 | Question 26

How many input assignments does an implicant fixing 2 of 5 Boolean inputs cover?

- 1. 4
- 2. 32
- 3. 2
- 4. 8

Week 04 | Question 27

Which pair of binary minterms can be directly combined by removing one differing literal?

- 1. 010 and 101.
- 2. 010 and 011.
- 3. 001 and 110.
- 4. 000 and 111.

Week 04 | Question 28

What is a typical limitation of heuristic logic minimization?

- 1. It applies only after chip packaging.
- 2. It requires every input assignment to be an OFF-set point.
- 3. It need not guarantee a globally minimum cover for every problem.
- 4. It can never reduce an expression.

Week 04 | Question 29

What does logic equivalence preservation require?

- 1. The transformed circuit matches the required behavior over the specified domain.
- 2. All physical wires must retain identical lengths.
- 3. Every internal signal must retain its old name.
- 4. The source text must remain character-for-character identical.

Week 04 | Question 30

Why can two equivalent Boolean expressions have different implementation quality?

- 1. Equivalence requires exactly the same cell area.
- 2. All equivalent expressions must have identical power.
- 3. Library choices, fanout, depth and interconnect can differ.
- 4. The truth table determines every physical coordinate uniquely.

Week 04 | Question 31

What often improves when a deep logic chain is restructured into a shallower balanced tree?

- 1. The number of logic levels on long paths can decrease.
- 2. The need to preserve functionality disappears.
- 3. Hold checks become irrelevant.
- 4. Every input capacitance necessarily becomes zero.

Week 04 | Question 32

An 8-input reduction built as a balanced tree of two-input gates has how many gate levels?

- 1. 2
- 2. 3
- 3. 7
- 4. 8

Week 04 | Question 33

An 8-input reduction built as a serial chain of two-input gates has how many gate levels?

- 1. 3
- 2. 4
- 3. 8
- 4. 7

Week 04 | Question 34

Which optimization can remove logic that cannot influence any required output or state?

- 1. Via redundancy insertion.
- 2. Clock-tree balancing.
- 3. Wafer probing.
- 4. Dead-logic elimination.

Week 04 | Question 35

Why should simulation models used as references handle reset and unknown values deliberately?

- 1. Unknown values are automatically identical to zero in every operator.
- 2. Synthesis never observes any constant.
- 3. Initialization and four-state behavior can affect the comparison.
- 4. Reset is only a packaging property.

Week 04 | Question 36

Which construct is generally a testbench timing control rather than portable ASIC synthesizable RTL?

- 1. A combinational AND expression.
- 2. An edge-triggered register update.
- 3. A procedural # delay used to model elapsed simulation time.
- 4. A static parameter width.

Week 04 | Question 37

Why can division by a known power of two simplify for unsigned arithmetic?

- 1. It changes unsigned addition into subtraction.
- 2. It can map to a right shift with appropriate width semantics.
- 3. It must add an extra clock domain.
- 4. It always requires a floating-point CPU.

Week 04 | Question 38

What is 8-bit unsigned 40 divided by 4?

- 1. 160
- 2. 8
- 3. 10
- 4. 20

Week 04 | Question 39

Which tool is commonly used to compile and simulate Verilog in this course?

- 1. Icarus Verilog.
- 2. A detailed router only.
- 3. A wafer prober.
- 4. GTKWave alone.

Week 04 | Question 40

What is GTKWave mainly used for?

- 1. Inspecting recorded signal waveforms.
- 2. Replacing an RTL compiler in every flow.
- 3. Computing production sales tax.
- 4. Performing transistor fabrication.

Week 04 | Question 41

What does a coverage-analysis tool add to a simulation workflow?

- 1. Detailed routing for every net.
- 2. Measurements of which modeled structures or conditions the tests exercised.
- 3. An automatic manufacturing-yield guarantee.
- 4. A proof that no functional requirement was omitted.

Week 04 | Question 42

Why must the testbench and DUT port connections be checked?

- 1. Every simulator repairs mismatched connections automatically.
- 2. Port names are never used during elaboration.
- 3. A correct test algorithm can still drive or observe the wrong signal.
- 4. Inputs and outputs are interchangeable without consequences.

Week 04 | Question 43

What can happen if a combinational sensitivity list omits an input read by the block?

- 1. Every gate doubles in physical size.
- 2. The omitted input becomes an ideal clock.
- 3. The fabrication process changes.
- 4. Simulation may fail to update when that input changes.

Week 04 | Question 44

What does always @* help express in traditional Verilog?

- 1. A mandatory asynchronous reset.
- 2. An unconditional fabrication step.
- 3. A block that executes only at the first clock edge.
- 4. A combinational sensitivity set inferred from referenced signals.

Week 04 | Question 45

What is a priority structure in an if/else-if chain?

- 1. Earlier satisfied conditions take precedence over later ones.
- 2. Every branch must execute simultaneously.
- 3. The last condition always wins regardless of earlier conditions.
- 4. All conditions are ignored after synthesis.

Week 04 | Question 46

For a two-input mux y=sel?a:b, what input determines y when sel=0?

- 1. a
- 2. b
- 3. sel itself always
- 4. The old value of y necessarily

Week 04 | Question 47

Consider synthesis don't-cares. A. They allow implementation freedom in the stated domain. B. They guarantee a stable physical x voltage. C. They require care when comparing simulation assumptions with optimized hardware. Which statements are correct?

- 1. B and C only
- 2. A only
- 3. A, B and C
- 4. A and C only

Week 04 | Question 48

Consider equivalent combinational circuits. A. Required output values agree for the same allowed inputs. B. Internal names must be identical. C. Cell counts can differ. Which statements are correct?

- 1. A and B only
- 2. B and C only
- 3. A and C only
- 4. A, B and C

Week 04 | Question 49

What is the minimum number of binary input bits needed to select among 8 choices?

1. 8
2. 3
3. 2
4. 4

Week 04 | Question 50

Which metric should accompany gate-count comparison when timing matters?

1. Critical-path delay under the stated implementation assumptions.
2. Only the number of comments.
3. Only the order of module names.
4. Only the source file creation date.

Week 05 - Multi-level optimization, FSMs, BDDs and SAT

50 practice MCQs | one point per question | no negative marking in this practice set

Week 05 | Question 01

For Boolean factoring, consider ab+ac=a(b+c). A. The two forms have the same truth table. B. Their literal-occurrence counts are necessarily equal. C. Their physical timing need not be equal. Which statements are correct?

- 1. B only
- 2. A and B only
- 3. A and C only
- 4. A, B and C

Week 05 | Question 02

Consider common-subexpression extraction. A. It can reduce duplicated logic. B. It can increase a shared node's fanout. C. It always improves the critical path. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. A and C only
- 4. B and C only

Week 05 | Question 03

Consider sharing and timing. A. Shared logic can need buffering for added load. B. Logical equivalence fixes every cell's electrical environment. C. A lower gate count alone does not prove a shorter critical path. Which statements are correct?

- 1. A and B only
- 2. B only
- 3. A and C only
- 4. A, B and C

Week 05 | Question 04

What is the main objective of multi-level logic optimization?

- 1. Improve a network with multiple logic stages while preserving required behavior.
- 2. Replace all sequential behavior with unknown values.
- 3. Only reduce package cost after fabrication.
- 4. Force all functions into one fixed two-level implementation.

Week 05 | Question 05

What is an observability don't-care for an internal node?

- 1. Any input combination that changes an output.
- 2. A value that must be represented as high impedance.
- 3. A condition under which changing that node does not affect observed outputs.
- 4. Every unreachable state regardless of its use assumptions.

Week 05 | Question 06

For y=n AND c, under which condition is n unobservable at y in a two-state Boolean model?

- 1. c=1
- 2. n=0 only
- 3. c=0
- 4. n=1 only

Week 05 | Question 07

For y=n OR c, under which condition is n unobservable at y in a two-state Boolean model?

- 1. c=0
- 2. c=1
- 3. n=0 only
- 4. n=1 only

Week 05 | Question 08

What is required before treating an unreachable-state condition as a valid don't-care?

- 1. Assume every inconvenient state is impossible.
- 2. Justify reachability from the allowed initial states and environment.
- 3. Remove reset verification from the design.
- 4. Check only the alphabetical order of state names.

Week 05 | Question 09

What describes a finite state machine?

- 1. A set of fabrication masks with no logical behavior.
- 2. Only a combinational truth table without any state.
- 3. Only an operating-system process list.
- 4. State storage plus next-state and output behavior.

Week 05 | Question 10

What distinguishes a Moore output from a Mealy output in the usual model?

- 1. Both outputs must depend only on the clock period.
- 2. A Mealy machine cannot have registers.
- 3. A Moore machine cannot have any inputs.
- 4. A Moore output depends on current state; a Mealy output may also depend on current inputs.

Week 05 | Question 11

What is a necessary notion for merging equivalent states?

- 1. They occupy adjacent physical locations.
- 2. Their names have the same length.
- 3. Their current output is the same on one sampled input only.
- 4. Their observable future behavior matches for all allowed future input sequences.

Week 05 | Question 12

What is the minimum binary state width for 10 states?

- 1. 5 bits
- 2. 10 bits
- 3. 3 bits
- 4. 4 bits

Week 05 | Question 13

How many state bits does a direct one-hot encoding of 10 states use?

- 1. 10 bits
- 2. 100 bits
- 3. 3 bits
- 4. 4 bits

Week 05 | Question 14

With 3 binary state bits, how many distinct encodings are available?

- 1. 3
- 2. 6
- 3. 8
- 4. 9

Week 05 | Question 15

If an FSM has 7 reachable states and 2 unreachable states, how many remain after removing only the unreachable states?

- 1. 5
- 2. 7
- 3. 9
- 4. 2

Week 05 | Question 16

Why can one-hot encoding improve some FPGA FSM timing?

- 1. It can simplify state decoding at the cost of more state bits.
- 2. It removes all reset requirements.
- 3. It always uses fewer flip-flops than binary encoding.
- 4. It guarantees fewer wires under every placement.

Week 05 | Question 17

Which action is an FSM optimization?

- 1. Replacing every state with an unrelated random value.
- 2. Skipping verification of the transformed machine.
- 3. Removing unreachable states under justified assumptions.
- 4. Changing every output to a constant regardless of the specification.

Week 05 | Question 18

Why must reset behavior be included when comparing two FSMs?

- 1. All state encodings reset to every state simultaneously.
- 2. It determines the allowed initial state or state correspondence.
- 3. Reset is irrelevant to reachability.
- 4. Reset affects only source-code indentation.

Week 05 | Question 19

What is the Boolean satisfiability (SAT) problem?

- 1. Determine whether some assignment makes a Boolean formula true.
- 2. Count only the number of transistors in a netlist.
- 3. Measure analog current without a model.
- 4. Decide a chip package material.

Week 05 | Question 20

Which formula is unsatisfiable?

- 1. x AND NOT x.
- 2. x OR NOT x.
- 3. x OR y.
- 4. x AND y.

Week 05 | Question 21

Which formula is a tautology?

- 1. x OR NOT x.
- 2. x XOR x.
- 3. x AND y.
- 4. x AND NOT x.

Week 05 | Question 22

How many satisfying assignments does x OR y have over two Boolean inputs?

- 1. 3
- 2. 2
- 3. 1
- 4. 4

Week 05 | Question 23

How many satisfying assignments does x AND y have over two Boolean inputs?

- 1. 2
- 2. 0
- 3. 3
- 4. 1

Week 05 | Question 24

What is a clause in conjunctive normal form?

- 1. A single mandatory state register.
- 2. A product of all the clauses.
- 3. A physical timing arc.
- 4. A disjunction of literals.

Week 05 | Question 25

What is a CNF formula?

- 1. A conjunction of clauses, with each clause a disjunction of literals.
- 2. An arbitrary waveform screenshot.
- 3. A gate library without Boolean functions.
- 4. A list containing only integer delays.

Week 05 | Question 26

For clauses (x) AND (NOT x OR y), what does unit propagation force?

- 1. No variable can be forced.
- 2. x=1 and then y=1.
- 3. x=0 and y=0.
- 4. x=1 and y=0.

Week 05 | Question 27

What is a SAT counterexample in a combinational equivalence check?

- 1. A proof that every implementation is physically legal.
- 2. An input assignment on which the compared circuits differ.
- 3. A random state name unrelated to an output.
- 4. A report that there are no source files.

Week 05 | Question 28

What is a binary decision diagram (BDD) used to represent?

- 1. A Boolean function through variable tests and branches.
- 2. A file-system directory tree with no logic meaning.
- 3. Only package-pin capacitance.
- 4. Only physical wire geometry.

Week 05 | Question 29

For a BDD decision node, what do the two outgoing branches conventionally mean?

- 1. The selected Boolean variable is 0 or 1.
- 2. A transistor is manufactured in two different wafers.
- 3. The clock is slow or fast.
- 4. Two routing layers must be connected.

Week 05 | Question 30

What is an ordered BDD?

- 1. One whose paths respect a fixed variable order.
- 2. One with no terminal nodes.
- 3. One whose nodes are sorted by physical cell area.
- 4. One containing only OR gates.

Week 05 | Question 31

With a fixed variable ordering, what useful property does an ROBDD have?

- 1. A size that is always linear in the number of variables.
- 2. A guaranteed minimum physical wire length.
- 3. A canonical representation of a Boolean function.
- 4. Identical size for all variable orderings.

Week 05 | Question 32

Which reduction removes a redundant decision node in a BDD?

- 1. Duplicate it whenever its successors are identical.
- 2. Force a new variable order on only one branch.
- 3. Replace all terminal 1 nodes with 0.
- 4. Delete the node when its low and high successors are identical.

Week 05 | Question 33

Which other reduction is standard for ROBDDs?

- 1. Merge nodes with the same variable and identical low/high successors.
- 2. Swap terminal meanings on every other level.
- 3. Merge arbitrary nodes merely because they have the same drawing color.
- 4. Delete all nodes representing the first variable.

Week 05 | Question 34

Why does variable ordering matter for BDDs?

- 1. It changes the mathematical truth table of the specified function.
- 2. It can cause a large change in representation size.
- 3. It eliminates the need for terminal nodes.
- 4. It guarantees identical memory use for every function.

Week 05 | Question 35

What are the terminal nodes of a Boolean BDD?

- 1. The largest and smallest physical delays.
- 2. The constants 0 and 1.
- 3. The first and last source-code lines.
- 4. The chip's input and output packages.

Week 05 | Question 36

If two functions have identical ROBDDs under the same order and conventions, what follows?

- 1. They must have the same original source text.
- 2. They must consume equal power at every voltage.
- 3. They must have identical transistor layouts.
- 4. They are logically equivalent.

Week 05 | Question 37

What is a formal property?

- 1. A precise statement of required behavior checked against a model.
- 2. Only a statement that the design is complicated.
- 3. A mandatory waveform color.
- 4. The date of a synthesis run.

Week 05 | Question 38

What distinguishes exhaustive formal reasoning from finite random testing?

- 1. Formal proofs automatically include every fabrication defect.
- 2. Formal methods can reason over all behaviors within the model and assumptions.
- 3. Random testing necessarily covers every possible sequence.
- 4. Formal methods need no model or assumptions.

Week 05 | Question 39

Why are environment assumptions significant in formal verification?

- 1. They can never remove any behavior from the model.
- 2. They prove the assumptions themselves are valid in every system.
- 3. They only change the PDF report format.
- 4. They define which input behaviors the proof covers.

Week 05 | Question 40

What risk results from overconstraining a formal environment?

- 1. All unreachable states become reachable automatically.
- 2. Realistic failing behaviors may be excluded from the proof.
- 3. The design is guaranteed to become slower.
- 4. The solver must always report a syntax error.

Week 05 | Question 41

What does an abstraction in formal analysis usually do?

- 1. Simplify the model while preserving the relevant property relationship.
- 2. Replace all specified behavior with arbitrary outputs without qualification.
- 3. Add detailed physical geometry to every Boolean variable.
- 4. Eliminate all need to interpret a proof result.

Week 05 | Question 42

Why should a reported counterexample be inspected?

- 1. Counterexamples cannot include input sequences.
- 2. Every counterexample proves the tool itself is incorrect.
- 3. Counterexamples always describe manufacturing defects.
- 4. It may expose a design bug, a specification mismatch or an assumption issue.

Week 05 | Question 43

What does SAT on a mismatch miter indicate?

- 1. At least one allowed input makes an output differ.
- 2. No allowed input can cause a mismatch.
- 3. The layout is DRC-clean.
- 4. The timing constraints are complete.

Week 05 | Question 44

A complete combinational mismatch miter is UNSAT. A. No modeled input causes an output mismatch. B. All physical timing requirements are thereby proved. C. The result depends on the comparison domain and modeling assumptions. Which statements are correct?

- 1. A only
- 2. B and C only
- 3. A, B and C
- 4. A and C only

Week 05 | Question 45

Consider the open-source tools in this course. A. Yosys performs RTL synthesis. B. GTKWave is mainly a waveform viewer. C. A waveform viewer alone performs complete RTL synthesis. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. B only
- 4. A and C only

Week 05 | Question 46

Why run optimization after converting processes to logic in a synthesis flow?

- 1. To replace all constraints with random values.
- 2. To force every signal to remain unknown.
- 3. To manufacture the final masks directly.
- 4. To simplify and transform the inferred network before later implementation steps.

Week 05 | Question 47

What does equivalence checking after optimization help detect?

- 1. Every possible thermal reliability failure.
- 2. Only the price of a cell library.
- 3. Only missing source-code comments.
- 4. Accidental changes in required functional behavior.

Week 05 | Question 48

How many distinct states can 5 binary flip-flops encode?

- 1. 10
- 2. 32
- 3. 25
- 4. 5

Week 05 | Question 49

If one-hot encoding uses 12 bits and binary encoding uses the minimum for 12 states, how many extra state bits does one-hot use?

- 1. 8
- 2. 4
- 3. 9
- 4. 12

Week 05 | Question 50

Which is the soundest interpretation of a successful proof?

- 1. The design must have minimum area.
- 2. The stated property holds for the analyzed model under its assumptions.
- 3. All future changes to the RTL are automatically proved.
- 4. Every unspecified property of the manufactured chip is guaranteed.

Week 06 - Formal verification, equivalence and cell libraries

50 practice MCQs | one point per question | no negative marking in this practice set

Week 06 | Question 01

Consider model checking. A. It checks properties of a transition model. B. Assumptions can restrict analyzed behaviors. C. A successful modeled proof automatically covers every manufacturing defect. Which statements are correct?

- 1. B and C only
- 2. A and B only
- 3. A, B and C
- 4. A only

Week 06 | Question 02

Consider reachability. A. Initial-state assumptions matter. B. Every representable binary encoding must be reachable. C. Transitions determine which later states are attainable. Which statements are correct?

- 1. B only
- 2. A and B only
- 3. A and C only
- 4. A, B and C

Week 06 | Question 03

Consider bounded model checking at depth k. A. It can find a counterexample within the bound. B. A clean bounded result alone always proves arbitrary-depth safety. C. A completeness argument is needed to justify an unbounded conclusion from a bound. Which statements are correct?

- 1. A, B and C
- 2. A and B only
- 3. B only
- 4. A and C only

Week 06 | Question 04

An assertion holds for all executions up to depth 5. What can be concluded without further reasoning?

- 1. No violation was found within that bound.
- 2. The property is proved for every execution length.
- 3. The design has minimum area.
- 4. Every reset state is reachable from every other state.

Week 06 | Question 05

What is the role of induction in an unbounded safety proof?

- 1. Establish an initial condition and show the property is preserved by transitions.
- 2. Replace the state-transition relation with a timing report.
- 3. Ignore all initial conditions.
- 4. Enumerate only one arbitrary test vector.

Week 06 | Question 06

What does a combinational equivalence check compare?

- 1. The input-output functions of two combinational representations under the specified domain.
- 2. Only their source-code line counts.
- 3. Their power dissipation at every future process node.
- 4. Only their placed cell coordinates.

Week 06 | Question 07

How is a mismatch miter commonly formed?

- 1. XOR corresponding outputs and OR the differences.
- 2. AND every input with ground.
- 3. Delete every output from both designs.
- 4. Connect all outputs directly together without comparison.

Week 06 | Question 08

For two 4-output circuits, how many pairwise XOR comparisons are needed in a simple mismatch miter?

- 1. 8
- 2. 1
- 3. 4
- 4. 16

Week 06 | Question 09

If the miter output can be 1, what does that demonstrate?

- 1. A counterexample to equivalence exists in the modeled domain.
- 2. The functions agree on every input.
- 3. Both netlists are physically legal.
- 4. Both designs have zero delay.

Week 06 | Question 10

Which assumption is important when using partitioned equivalence checking?

- 1. Every internal signal name proves equivalence automatically.
- 2. Correspondence and boundary assumptions must correctly connect the compared partitions.
- 3. Partitions must have equal physical areas.
- 4. Arbitrary partition inputs can be ignored without justification.

Week 06 | Question 11

Why can black-boxing affect a proof's meaning?

- 1. A black box automatically behaves like every possible real implementation in the same way.
- 2. It resolves all timing constraints.
- 3. It removes all design inputs.
- 4. The proof depends on what behavior is assumed for the omitted block.

Week 06 | Question 12

What does a Liberty technology library normally provide?

- 1. Cell logic and characterized electrical/timing information.
- 2. Only the finished GDS layout of the top-level design.
- 3. The complete functional specification of every customer chip.
- 4. Only fabrication defect measurements.

Week 06 | Question 13

What is a timing arc?

- 1. A package outline drawing.
- 2. A manufacturing wafer boundary.
- 3. A file-system symbolic link.
- 4. A modeled timing relationship from a related pin or event to another pin.

Week 06 | Question 14

For an inverter's negative-unate arc, an input rise normally corresponds to what output transition?

- 1. A rise necessarily.
- 2. No possible transition.
- 3. A change in clock frequency only.
- 4. A fall.

Week 06 | Question 15

For an AND gate with other inputs held at noncontrolling values, the input-to-output arc is usually what?

- 1. A sequential setup constraint only.
- 2. Always non-unate.
- 3. Positive unate.
- 4. Negative unate.

Week 06 | Question 16

Why can an XOR input-to-output relationship be non-unate?

- 1. XOR has no input pins.
- 2. The output never changes.
- 3. The output transition direction depends on the other input's value.
- 4. Every input rise always causes an output rise.

Week 06 | Question 17

What does a basic NLDM propagation-delay table commonly use as indices?

- 1. Clock period and the number of comments.
- 2. Input transition time and output load capacitance.
- 3. Only package weight and die color.
- 4. Only source-code width and file size.

Week 06 | Question 18

What does an output transition table model?

- 1. The distance between two module source files.
- 2. The yield of a wafer after dicing.
- 3. Only the clock's absolute arrival timestamp.
- 4. Output slew under the specified input slew and load conditions.

Week 06 | Question 19

What is slew?

- 1. The time between two unrelated chip deliveries.
- 2. The count of gates in a truth table.
- 3. The total number of clock cycles since reset only.
- 4. The time taken for a signal transition between specified voltage thresholds.

Week 06 | Question 20

Why should slew thresholds be specified?

- 1. Thresholds determine the number of RTL ports.
- 2. A transition time depends on which voltage levels define its measurement interval.
- 3. Voltage levels never affect transition-time definitions.
- 4. Every library uses an identical threshold convention by necessity.

Week 06 | Question 21

An inverter input crossing occurs at 10 ns and its corresponding output crossing at 10.08 ns. What is the propagation delay?

- 1. 20.08 ns
- 2. 0.8 ns
- 3. 80 ps
- 4. 8 ps

Week 06 | Question 22

At a fixed input slew, a table gives delay 20 ps at 10 fF and 40 ps at 30 fF. What is linear-interpolated delay at 20 fF?

- 1. 60 ps
- 2. 30 ps
- 3. 40 ps
- 4. 25 ps

Week 06 | Question 23

For a table f(x,y) with corner values 10,20,30,40, what is its bilinear value at the center of that rectangular cell?

- 1. 30
- 2. 25
- 3. 100
- 4. 20

Week 06 | Question 24

Why must values outside a characterized table's range be treated carefully?

- 1. The logic function of the cell always changes.
- 2. Results may depend on the tool's extrapolation or limiting behavior.
- 3. All tools necessarily use identical extrapolation.
- 4. Every out-of-range value is exactly zero.

Week 06 | Question 25

What is setup time relative to a capture clock edge?

- 1. The wire delay from one package to another.
- 2. The RTL compilation duration.
- 3. The required interval during which data must be stable before that edge.
- 4. The minimum time the clock must remain low after power-off.

Week 06 | Question 26

What is hold time relative to a capture edge?

- 1. The number of registers in a pipeline.
- 2. Only the maximum combinational delay.
- 3. The entire clock period of the chip.
- 4. The required interval during which data must remain stable after that edge.

Week 06 | Question 27

What does clock-to-Q delay measure for a flip-flop?

- 1. The period of an arbitrary unrelated clock.
- 2. Delay from the active clock event to the corresponding Q transition.
- 3. Delay from the first source-code line to the last.
- 4. The physical length of the clock-tree source file.

Week 06 | Question 28

Which values commonly index a sequential setup or hold constraint table?

- 1. Data transition and related clock transition.
- 2. Only a testbench's total simulation duration.
- 3. Only die width and die height.
- 4. Only output net name and instance name.

Week 06 | Question 29

In a fixed library corner, why can two flip-flop instances have different clock-to-Q delays?

- 1. Every instance has the exact same electrical environment.
- 2. The Boolean name of D automatically changes its function.
- 3. Their clock slew and output loading can differ.
- 4. Placement never affects loading.

Week 06 | Question 30

What does a process-voltage-temperature corner specify?

- 1. A list of package colors.
- 2. Three arbitrary clock phase names.
- 3. A modeled fabrication condition, supply condition and temperature.
- 4. The chip's three geometric corner coordinates.

Week 06 | Question 31

Why should signoff consider more than one operating corner?

- 1. A design has no variation after synthesis.
- 2. One corner always proves every physical effect in all conditions.
- 3. Corners are used only to render tables.
- 4. Different conditions can stress different maximum and minimum timing checks.

Week 06 | Question 32

Why is a slow-corner setup check insufficient to establish hold timing?

- 1. Hold is a maximum-delay-only requirement.
- 2. Setup and hold are mathematically identical checks.
- 3. A slower clock always repairs hold.
- 4. Hold depends on minimum data-path behavior and clock relationships as well.

Week 06 | Question 33

What information is associated with a sequential cell's functional model?

- 1. Its state-update and output behavior.
- 2. Only a printable logo.
- 3. Every future routed net's capacitance.
- 4. The customer's complete floorplan.

Week 06 | Question 34

What is input pin capacitance used for in timing analysis?

- 1. Setting every net delay to zero.
- 2. Modeling the load that the pin presents to its driver.
- 3. Replacing the Boolean function of the receiving cell.
- 4. Recording the number of source statements.

Week 06 | Question 35

Two receiver pins have 2 fF and 3 fF capacitance, and the wire adds 4 fF. What is the simple total capacitive load?

- 1. 24 fF
- 2. 5 fF
- 3. 9 fF
- 4. 4 fF

Week 06 | Question 36

How is cell delay different from interconnect delay?

- 1. The two quantities have different units of time by definition.
- 2. Cell delay never depends on loading.
- 3. Interconnect delay is always exactly zero.
- 4. Cell delay is through a cell arc; interconnect delay is through the connecting net model.

Week 06 | Question 37

What does a max-transition constraint limit?

- 1. Only the number of cell instances in a design.
- 2. The number of legal state encodings.
- 3. The magnitude of a Verilog integer constant.
- 4. The allowed signal transition time.

Week 06 | Question 38

What does a max-capacitance constraint limit?

- 1. The number of proof clauses in a SAT solver.
- 2. The capacitive load a driver is allowed to see.
- 3. The distance between file directories.
- 4. The clock period in all designs.

Week 06 | Question 39

Why can a functionally equivalent replacement violate an electrical constraint?

- 1. Every equivalent gate has the same transistor sizes.
- 2. Logical equivalence guarantees identical input capacitance.
- 3. Electrical constraints are unrelated to gates.
- 4. Equivalent Boolean behavior does not ensure adequate drive or acceptable slew/loading.

Week 06 | Question 40

What is a conditional timing arc?

- 1. A route existing only in a PDF illustration.
- 2. A delay measured only when the tool's user is logged in.
- 3. An arc that requires no related pin.
- 4. An arc whose applicability depends on a specified logical condition.

Week 06 | Question 41

If a cell's input slew worsens while load is fixed, which conclusion is safest without its table?

- 1. The delay must become exactly zero.
- 2. The Boolean function must invert.
- 3. The delay must halve for every cell.
- 4. Consult the characterized table rather than assume an exact delay change.

Week 06 | Question 42

Which sequence is appropriate for a simple CEC flow?

- 1. Compare only file creation dates.
- 2. Delete outputs, ignore inputs, then claim equivalence.
- 3. Prepare corresponding designs, create a mismatch comparison, then prove or find a counterexample.
- 4. Place cells, package chips, then write the specification.

Week 06 | Question 43

What property can a counterexample to a safety assertion demonstrate?

- 1. A reachable execution violating the assertion under the modeled assumptions.
- 2. That every possible execution violates it.
- 3. That the design has no outputs.
- 4. That all technology libraries are invalid.

Week 06 | Question 44

An FSM reset state is S0, and only transitions S0 to S1 and S1 to S1 exist. Which listed state is unreachable?

- 1. Both S0 and S1
- 2. S1
- 3. S0
- 4. S2

Week 06 | Question 45

Why can state-space size grow rapidly with register count?

- 1. Only the output port count matters.
- 2. n binary state bits admit up to 2 to the power n encodings.
- 3. State encodings grow only as n+1.
- 4. Every register removes one possible state.

Week 06 | Question 46

How many binary encodings can 8 state bits represent?

- 1. 64
- 2. 8
- 3. 16
- 4. 256

Week 06 | Question 47

Consider library-based timing abstraction. A. Characterization supports efficient delay estimation. B. The models have operating conditions and ranges. C. They guarantee exact behavior in every unmodeled analog situation. Which statements are correct?

- 1. A, B and C
- 2. A and B only
- 3. A only
- 4. B and C only

Week 06 | Question 48

Consider library units. A. A numerical delay must be read with its time unit. B. Capacitance units affect interpretation of loads. C. Signal names automatically correct unit mismatches. Which statements are correct?

- 1. A, B and C
- 2. B only
- 3. A and B only
- 4. A and C only

Week 06 | Question 49

A library uses a 1 ns time unit. What physical delay does a value of 0.025 represent?

- 1. 250 ps
- 2. 25 ps
- 3. 0.025 ps
- 4. 25 ns

Week 06 | Question 50

What does a successful library-based implementation still require?

- 1. Replacing all registers by constants.
- 2. No checks after mapping.
- 3. Verification against functional, timing and physical requirements in the intended context.
- 4. A guarantee that every fabricated die passes.

Week 07 - Static timing analysis and OpenSTA

50 practice MCQs | one point per question | no negative marking in this practice set

Week 07 | Question 01

Consider ordinary synchronous setup analysis. A. It normally uses late/max data arrival. B. It protects data arrival before the intended capture edge. C. It proves every hold check automatically. Which statements are correct?

- 1. B and C only
- 2. A and B only
- 3. A, B and C
- 4. A only

Week 07 | Question 02

Consider same-edge hold analysis. A. It is concerned with data changing too early. B. It normally uses early/minimum data delay. C. Increasing the clock period always repairs a violation. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. A and C only
- 4. B only

Week 07 | Question 03

Consider an ordinary register-to-register setup budget. A. Launch clock-to-Q delay contributes. B. Capture setup time consumes available margin. C. Only combinational delay is needed to determine the period. Which statements are correct?

- 1. A only
- 2. A, B and C
- 3. B and C only
- 4. A and B only

Week 07 | Question 04

Ignoring skew and uncertainty, which hold inequality is appropriate?

- 1. T is always equal to tHold.
- 2. tCQ_max + d_max + tSetup is at most zero.
- 3. tHold must exceed every possible data delay.
- 4. tCQ_min + d_min is at least tHold.

Week 07 | Question 05

For T=1000 ps, tCQ_max=80 ps, d_max=700 ps and tSetup=70 ps, with zero skew and uncertainty, what is setup slack?

- 1. 850 ps
- 2. -150 ps
- 3. 150 ps
- 4. 230 ps

Week 07 | Question 06

For tCQ_min=30 ps, d_min=20 ps and tHold=40 ps, with zero skew and uncertainty, what is hold slack?

- 1. -10 ps
- 2. 50 ps
- 3. 10 ps
- 4. 90 ps

Week 07 | Question 07

With zero skew and uncertainty, tCQ_max=60 ps, d_max=840 ps and tSetup=100 ps, what is the minimum setup-safe clock period in this model?

- 1. 840 ps
- 2. 900 ps
- 3. 160 ps
- 4. 1000 ps

Week 07 | Question 08

A minimum setup-safe period is 2 ns in the given model. What maximum clock frequency corresponds to it?

- 1. 50 MHz
- 2. 2 GHz
- 3. 500 MHz
- 4. 250 MHz

Week 07 | Question 09

Define skew k as capture-clock arrival minus launch-clock arrival. What does positive k do to the setup budget?

- 1. It removes clock-to-Q delay.
- 2. It increases the available single-cycle setup time.
- 3. It necessarily makes setup worse by exactly one period.
- 4. It has no effect on any timing check.

Week 07 | Question 10

With the same capture-minus-launch skew convention, what does positive k do to hold margin?

- 1. It always improves hold margin.
- 2. It reduces hold margin.
- 3. It makes hold independent of clock arrival.
- 4. It is equivalent to deleting every data buffer.

Week 07 | Question 11

For T=1000 ps, k=50 ps, tCQ_max=80 ps, d_max=700 ps and tSetup=70 ps, ignoring uncertainty, setup slack is what?

- 1. -200 ps
- 2. 100 ps
- 3. 150 ps
- 4. 200 ps

Week 07 | Question 12

For k=20 ps, tCQ_min=30 ps, d_min=20 ps and tHold=40 ps, ignoring uncertainty, hold slack is what?

- 1. -10 ps
- 2. 10 ps
- 3. 70 ps
- 4. 30 ps

Week 07 | Question 13

For T=1000 ps, k=-40 ps, tCQ_max=60 ps, d_max=800 ps and tSetup=70 ps, ignoring uncertainty, setup slack is what?

- 1. 110 ps
- 2. 30 ps
- 3. -30 ps
- 4. 70 ps

Week 07 | Question 14

If setup uncertainty is 30 ps and the nominal setup slack is 80 ps, what is slack after subtracting this uncertainty?

- 1. 50 ps
- 2. -110 ps
- 3. 110 ps
- 4. 80 ps

Week 07 | Question 15

Why can slowing the clock fail to repair a same-edge hold violation?

- 1. The hold inequality is primarily a minimum-delay and relative-clock requirement, not the next-period setup budget.
- 2. Clock-to-Q delay becomes zero at slower frequency.
- 3. Every hold check includes three full clock periods.
- 4. Hold depends only on the source file size.

Week 07 | Question 16

A path has hold slack -25 ps. A repair adds 40 ps to its minimum data delay, with no other changes. New hold slack is what?

- 1. -15 ps
- 2. 15 ps
- 3. -65 ps
- 4. 65 ps

Week 07 | Question 17

A path has setup slack 35 ps. A change adds 20 ps to its maximum data delay, with no other changes. New setup slack is what?

- 1. 55 ps
- 2. -15 ps
- 3. -55 ps
- 4. 15 ps

Week 07 | Question 18

Which data is important for STA?

- 1. Only a single simulation waveform image.
- 2. Only the chip package invoice.
- 3. Only the source-code comments.
- 4. Netlist, timing libraries and timing constraints, plus interconnect modeling appropriate to the stage.

Week 07 | Question 19

What does an arrival time represent?

- 1. Only the time a file was downloaded.
- 2. When a signal can reach a point relative to a timing reference.
- 3. The number of sinks on the clock net.
- 4. The number of Boolean minterms.

Week 07 | Question 20

What does required time represent in a setup check?

- 1. The earliest time the source file may be edited.
- 2. The total number of gates in a library.
- 3. A mandatory value of zero at every endpoint.
- 4. The latest allowable data-arrival boundary for that check.

Week 07 | Question 21

What is setup slack with arrival 720 ps and required time 800 ps?

- 1. -80 ps
- 2. 1520 ps
- 3. 80 ps
- 4. 720 ps

Week 07 | Question 22

What is setup slack with arrival 900 ps and required time 850 ps?

- 1. 1750 ps
- 2. 50 ps
- 3. -50 ps
- 4. -1750 ps

Week 07 | Question 23

For a hold comparison, earliest arrival is 90 ps and required minimum arrival is 65 ps. What is hold slack?

- 1. 65 ps
- 2. 25 ps
- 3. 155 ps
- 4. -25 ps

Week 07 | Question 24

At a reconvergent node, how are late arrivals commonly propagated for a maximum timing check?

- 1. Always take the arithmetic average of incoming arrivals.
- 2. Always select the shortest source-code name.
- 3. Take the maximum applicable incoming arrival plus its corresponding arc delay.
- 4. Add all incoming arrivals together regardless of path structure.

Week 07 | Question 25

At a reconvergent node, how are early arrivals commonly propagated for a minimum timing check?

- 1. Replace every early arrival with one clock period.
- 2. Take the largest incoming arrival regardless of arc.
- 3. Take the minimum applicable incoming arrival plus its corresponding arc delay.
- 4. Multiply all arrivals together.

Week 07 | Question 26

Two candidate late arrival-plus-arc values are 120 ps and 170 ps. What late arrival propagates?

- 1. 290 ps
- 2. 145 ps
- 3. 120 ps
- 4. 170 ps

Week 07 | Question 27

Two candidate early arrival-plus-arc values are 120 ps and 170 ps. What early arrival propagates?

- 1. 290 ps
- 2. 145 ps
- 3. 170 ps
- 4. 120 ps

Week 07 | Question 28

What can make graph-based analysis pessimistic compared with path-based analysis?

- 1. Always using exact joint path histories at no computational cost.
- 2. Combining conservative quantities that do not occur together on the same realizable path.
- 3. Considering only the newest source file.
- 4. Ignoring every load capacitance.

Week 07 | Question 29

What does path-based analysis attempt to retain more precisely?

- 1. An assumption that all paths are identical.
- 2. Only aggregate package price.
- 3. Only the number of RTL files.
- 4. The conditions and delay/slew behavior along a particular path.

Week 07 | Question 30

Why is a critical-path report useful?

- 1. It replaces the design specification.
- 2. It identifies a path limiting the stated timing requirement.
- 3. It proves all functional behaviors correct.
- 4. It guarantees zero routing congestion.

Week 07 | Question 31

What is a common data-path startpoint?

- 1. A launch register's clocked output or a constrained input port.
- 2. Any comment in a Verilog file.
- 3. A package label.
- 4. Only the edge of the wafer.

Week 07 | Question 32

What is a common data-path endpoint?

- 1. A capture register's data pin or a constrained output port.
- 2. The power supply's marketing name.
- 3. A tool installation directory.
- 4. Only an RTL module name.

Week 07 | Question 33

Why are rising and falling transitions often analyzed separately?

- 1. Rising transitions never use timing arcs.
- 2. A falling transition has no electrical meaning.
- 3. Both transition types always have identical delay.
- 4. Their cell and net delays can differ.

Week 07 | Question 34

Why should multiple process and environmental scenarios be analyzed?

- 1. Different scenarios can limit different paths or timing checks.
- 2. One arbitrary scenario necessarily bounds all checks.
- 3. A netlist can exist at only one temperature.
- 4. Scenarios affect only report typography.

Week 07 | Question 35

What does on-chip variation modeling address?

- 1. Delay variation among different paths or locations within one chip.
- 2. The spelling of clock names.
- 3. Only a source-file syntax variation.
- 4. Only the number of separate wafers ordered.

Week 07 | Question 36

What is a purpose of common-path pessimism removal in clock analysis?

- 1. Remove every real clock delay.
- 2. Convert all hold checks into setup checks.
- 3. Remove inconsistent early/late treatment of a physically shared clock path.
- 4. Ignore the capture clock entirely.

Week 07 | Question 37

A timing path has three combinational delays 40,60,80 ps. Ignoring net delays, what is their sum?

- 1. 100 ps
- 2. 240 ps
- 3. 80 ps
- 4. 180 ps

Week 07 | Question 38

Add launch tCQ=30 ps to combinational delay 180 ps. What is data arrival relative to the launch edge, ignoring other delays?

- 1. 180 ps
- 2. 210 ps
- 3. 150 ps
- 4. 30 ps

Week 07 | Question 39

If capture required time is 260 ps for the arrival in the preceding defined 210 ps case, what is setup slack?

- 1. 50 ps
- 2. 470 ps
- 3. 210 ps
- 4. -50 ps

Week 07 | Question 40

Two paths have slacks -40 ps and +15 ps. What is the worst slack?

- 1. -40 ps
- 2. -25 ps
- 3. +15 ps
- 4. +40 ps

Week 07 | Question 41

Three endpoint slacks are -30 ps, -10 ps and +20 ps. Summing only negative endpoint slacks gives what TNS?

- 1. -60 ps
- 2. -40 ps
- 3. -20 ps
- 4. +40 ps

Week 07 | Question 42

Which conclusion is justified by positive setup slack on one path?

- 1. Every operating mode is necessarily checked.
- 2. That particular modeled setup check meets its requirement.
- 3. Every hold path necessarily passes.
- 4. The chip has no functional errors.

Week 07 | Question 43

Why should an unconstrained path be investigated?

- 1. It always contains no gates.
- 2. It may be omitted from the intended timing checks.
- 3. It necessarily has infinite physical speed.
- 4. It proves that the design has no clocks.

Week 07 | Question 44

What is the role of an ideal-clock assumption in an early analysis?

- 1. Prove clock skew is physically impossible.
- 2. Delete all setup and hold checks.
- 3. Treat clock distribution according to an idealized model rather than a built physical clock network.
- 4. Set every data delay to zero.

Week 07 | Question 45

Why may post-route timing differ from synthesis timing?

- 1. Post-route analysis ignores libraries.
- 2. Routing cannot change load capacitance.
- 3. Synthesis and routing always use identical parasitics by definition.
- 4. Extracted interconnect and physical clock behavior can differ from earlier estimates.

Week 07 | Question 46

Which tool in the course performs static timing analysis?

- 1. A silicon dicing saw.
- 2. A shell text editor only.
- 3. OpenSTA.
- 4. GTKWave alone.

Week 07 | Question 47

Consider a timing-path report. A. The check type must be identified. B. Units and scenario must be read. C. A positive result on one path proves every design requirement. Which statements are correct?

- 1. A, B and C
- 2. A and C only
- 3. A and B only
- 4. B only

Week 07 | Question 48

For fixed path and clock quantities, consider increasing capture setup time. A. It reduces setup margin. B. It automatically removes clock skew. C. It does not make the combinational path delay vanish. Which statements are correct?

- 1. B and C only
- 2. A only
- 3. A and C only
- 4. A, B and C

Week 07 | Question 49

What is the hold impact of increasing tHold by 10 ps with other quantities fixed?

- 1. Setup period decreases by exactly 10 ps.
- 2. Clock-to-Q delay is removed.
- 3. Hold slack decreases by 10 ps.
- 4. Hold slack increases by 10 ps.

Week 07 | Question 50

Which interpretation of a timing report is sound?

- 1. A positive number alone proves every design requirement.
- 2. Read path type, units, scenario and constraints before interpreting the slack value.
- 3. Net names determine whether a path is physically slow.
- 4. All timing tools use the same unstated units in every library.

Week 08 - Constraints, technology mapping and timing optimization

50 practice MCQs | one point per question | no negative marking in this practice set

Week 08 | Question 01

Consider an SDC clock definition. A. It establishes a timing reference. B. It can describe the clock waveform. C. It physically builds the complete clock network by itself. Which statements are correct?

- 1. B and C only
- 2. A only
- 3. A and B only
- 4. A, B and C

Week 08 | Question 02

Consider create_clock. A. It is a common SDC command. B. Its period affects synchronous timing requirements. C. It proves clock-domain crossing correctness by itself. Which statements are correct?

- 1. A and B only
- 2. B only
- 3. A and C only
- 4. A, B and C

Week 08 | Question 03

A primary clock has period 8 ns. What is its frequency?

- 1. 62.5 MHz
- 2. 8 MHz
- 3. 800 MHz
- 4. 125 MHz

Week 08 | Question 04

For a 10 ns clock with waveform edges at 0 ns and 4 ns, what is the high-time fraction?

- 1. 40%
- 2. 60%
- 3. 50%
- 4. 25%

Week 08 | Question 05

Why might a generated clock need an explicit relationship to a source clock?

- 1. Generated clocks are only labels in simulation screenshots.
- 2. Generated clocks cannot have a period.
- 3. Every generated clock is necessarily asynchronous.
- 4. Its divide, multiply or edge relationship affects timing between domains.

Week 08 | Question 06

A generated clock divides a 200 MHz source by 4. What is its frequency?

- 1. 100 MHz
- 2. 800 MHz
- 3. 50 MHz
- 4. 25 MHz

Week 08 | Question 07

Consider input delays. A. They model external arrival assumptions. B. Min and max bounds can serve different timing checks. C. They necessarily insert physical delay cells at all input ports. Which statements are correct?

- 1. A only
- 2. A, B and C
- 3. B and C only
- 4. A and B only

Week 08 | Question 08

What does set_output_delay model?

- 1. A mandatory clock-tree insertion delay.
- 2. The number of output pins in the package.
- 3. Only the source file's save duration.
- 4. The external timing requirement imposed beyond a design output.

Week 08 | Question 09

Why are both max and min input delays useful?

- 1. Min values apply only to source-code comments.
- 2. Setup and hold require different external timing bounds.
- 3. Max and min are always identical in every system.
- 4. Max values remove all capture requirements.

Week 08 | Question 10

What does set_clock_uncertainty represent?

- 1. A guarantee of exact physical clock delivery.
- 2. The number of possible Verilog state values.
- 3. The source-code compiler's warning count.
- 4. Timing margin for specified uncertainty effects in the analysis.

Week 08 | Question 11

Why distinguish source latency from clock-network latency?

- 1. Neither can affect timing.
- 2. They refer to different portions of clock delivery relative to the defined design boundary.
- 3. They are names for source-code indentation.
- 4. Both necessarily equal the full data-path delay.

Week 08 | Question 12

What does a false-path exception do?

- 1. Exclude specified paths from the selected timing checks.
- 2. Prove that the excluded path is functionally impossible by itself.
- 3. Automatically insert synchronizers.
- 4. Physically remove every gate on the path.

Week 08 | Question 13

What must support a false-path constraint?

- 1. A justified functional or timing-intent reason.
- 2. Merely a desire to hide an inconvenient violation.
- 3. The fact that its net name is long.
- 4. A missing testbench comment.

Week 08 | Question 14

What does a multicycle setup exception normally change?

- 1. Which capture edge is used for the setup requirement of the specified path.
- 2. The number of metal layers in the process.
- 3. The physical oscillator frequency by itself.
- 4. The Boolean function of every gate.

Week 08 | Question 15

For a same-clock path intentionally allowed 3 setup cycles, why review the corresponding hold exception?

- 1. Hold analysis is automatically meaningless for all multicycle paths.
- 2. Multicycle paths contain no registers.
- 3. The setup edge shift can also change the hold relationship used by the tool.
- 4. Every hold requirement must become exactly zero.

Week 08 | Question 16

In the common same-clock convention, a setup multicycle of N is often paired with what hold adjustment to restore the original hold edge?

- 1. 2N
- 2. N+1
- 3. N-1
- 4. Zero in every possible case regardless of intent

Week 08 | Question 17

For a 5 ns clock, a correctly specified 3-cycle setup path has what ideal edge-to-edge budget before other deductions?

- 1. 20 ns
- 2. 10 ns
- 3. 15 ns
- 4. 5 ns

Week 08 | Question 18

Why should asynchronous clock domains be handled deliberately in timing constraints?

- 1. All crossings are safe without synchronization.
- 2. They always have identical edge times.
- 3. The timing tool creates every needed protocol automatically.
- 4. Their phase relationship is not a normal fixed synchronous launch/capture assumption.

Week 08 | Question 19

What is a risk of declaring clocks asynchronous too broadly?

- 1. Every data pin becomes a clock pin.
- 2. Every false path is automatically proved unreachable.
- 3. The chip's area necessarily halves.
- 4. Legitimate timing relationships may be omitted from analysis.

Week 08 | Question 20

What should happen to paths unintentionally left without clocks or I/O constraints?

- 1. Investigate and define the missing timing intent.
- 2. Assume they pass because no slack is printed.
- 3. Mark all of them as false paths without review.
- 4. Delete them from the functional specification.

Week 08 | Question 21

For an input-to-register path, max external input delay is 200 ps, internal max delay is 500 ps, setup is 80 ps and T=1000 ps. Ignore skew and uncertainty. Setup slack is what?

- 1. -220 ps
- 2. 220 ps
- 3. 420 ps
- 4. 300 ps

Week 08 | Question 22

For a register-to-output path, T=1000 ps, output required external delay is 150 ps, launch tCQ=70 ps and internal delay is 480 ps. Ignore other effects. Setup slack is what?

- 1. 450 ps
- 2. 300 ps
- 3. -300 ps
- 4. 550 ps

Week 08 | Question 23

An input-to-register early arrival is 40 ps external plus 30 ps internal. Required minimum arrival is 50 ps. What is hold slack?

- 1. 20 ps
- 2. -20 ps
- 3. 120 ps
- 4. 70 ps

Week 08 | Question 24

If setup uncertainty increases from 20 ps to 50 ps, with all else fixed, what happens to setup slack?

- 1. It increases by 30 ps.
- 2. It remains identical by definition.
- 3. It decreases by 70 ps.
- 4. It decreases by 30 ps.

Week 08 | Question 25

For a 10 ns clock, a same-domain falling-edge launch at 5 ns and next rising-edge capture at 10 ns provide what ideal edge budget?

- 1. 0 ns
- 2. 10 ns
- 3. 15 ns
- 4. 5 ns

Week 08 | Question 26

What is the aim of technology mapping?

- 1. Assign high-level algorithms to software licenses.
- 2. Determine wafer defects from an RTL file alone.
- 3. Implement a logical network with available target-library cells under constraints.
- 4. Replace every logic function by a wire.

Week 08 | Question 27

Why can different libraries give different PPA for the same Boolean network?

- 1. A mapped netlist does not depend on its target library.
- 2. Truth-table equivalence fixes every transistor size.
- 3. Available cell functions, sizes and characterized behavior differ.
- 4. All libraries contain identical cells at identical delays.

Week 08 | Question 28

What does increasing a cell's drive strength often trade?

- 1. Elimination of every input capacitance.
- 2. Better drive capability against increased area and input loading.
- 3. Better performance with guaranteed zero cost in every metric.
- 4. A changed Boolean truth table necessarily.

Week 08 | Question 29

Why can upsizing one gate hurt the preceding stage?

- 1. It removes the driver from the netlist automatically.
- 2. It turns the preceding stage into a package pin.
- 3. The larger gate can present more input capacitance to its driver.
- 4. The preceding stage can never see a load change.

Week 08 | Question 30

Which technique is often used to drive a heavily loaded net?

- 1. Changing the net's logic value arbitrarily.
- 2. Removing all required sinks.
- 3. Suppressing every timing report.
- 4. Buffer insertion or suitable driver sizing.

Week 08 | Question 31

Why may buffering improve a long interconnect's delay?

- 1. It proves the path functionally unreachable.
- 2. It removes all physical wire resistance.
- 3. A buffer has zero delay and zero area in silicon.
- 4. It can divide the electrical effort and isolate sections of the RC load.

Week 08 | Question 32

What is logic duplication used for in some timing optimizations?

- 1. Replace every sequential element by a constant.
- 2. Guarantee minimum total area.
- 3. Reduce load or fanout on a critical shared driver by creating separate equivalent drivers.
- 4. Change required output values.

Week 08 | Question 33

What is pin swapping in a timing context?

- 1. Swap a clock net with ground arbitrarily.
- 2. Reassign logically interchangeable input connections to exploit different pin timing/loading.
- 3. Exchange package serial numbers.
- 4. Change all input values to unknown.

Week 08 | Question 34

What does restructuring a logic network aim to change?

- 1. Only the font in the RTL editor.
- 2. The number of manufactured wafers only.
- 3. Its topology or depth while preserving the required function.
- 4. The product specification without review.

Week 08 | Question 35

What is the aim of sizing noncritical gates down, when constraints permit?

- 1. Force every path to fail setup.
- 2. Reduce area or power while retaining adequate timing margin.
- 3. Remove all timing analysis.
- 4. Increase every input capacitance.

Week 08 | Question 36

Which measure describes worst negative timing margin for a selected set of checks?

- 1. Total positive area.
- 2. Worst negative slack, under the report's convention.
- 3. The average source-code line length.
- 4. The number of clock definitions only.

Week 08 | Question 37

Which measure sums negative slack contributions over the report's selected endpoints or checks?

- 1. Maximum cell drive strength.
- 2. Peak clock frequency alone.
- 3. Total number of reset bits.
- 4. Total negative slack.

Week 08 | Question 38

Why recheck hold after a setup optimization?

- 1. Any setup improvement proves hold automatically.
- 2. Hold is a maximum-frequency-only test.
- 3. Hold matters only before synthesis.
- 4. Faster data paths or changed clocks can reduce minimum-delay margin.

Week 08 | Question 39

A setup repair removes 40 ps of max delay from a path whose setup slack was -25 ps. All else fixed, new slack is what?

- 1. -15 ps
- 2. 65 ps
- 3. -65 ps
- 4. 15 ps

Week 08 | Question 40

A hold repair adds 25 ps minimum delay to a path whose hold slack was -10 ps. All else fixed, new slack is what?

- 1. 35 ps
- 2. 15 ps
- 3. -15 ps
- 4. -35 ps

Week 08 | Question 41

A cell change adds 15 ps to max delay and 10 ps to min delay. Which direct effect follows with other values fixed?

- 1. Both checks necessarily fail regardless of their original margins.
- 2. Setup slack increases by 10 ps and hold slack decreases by 15 ps.
- 3. Setup slack decreases by 15 ps and hold slack increases by 10 ps.
- 4. Both setup and hold slack increase by 15 ps.

Week 08 | Question 42

Why must timing exceptions be reviewed after hierarchy or signal names change?

- 1. Their object queries may no longer target the intended paths.
- 2. Signal renaming removes all physical delays.
- 3. Every exception becomes a verified formal proof.
- 4. Constraint queries are guaranteed to follow every arbitrary text change.

Week 08 | Question 43

What does a get_ports query normally select?

- 1. All manufactured package labels.
- 2. Every internal transistor geometry.
- 3. Top-level design ports matching the query.
- 4. Only operating-system process IDs.

Week 08 | Question 44

What does a get_clocks query normally select?

- 1. Every net that happens to toggle.
- 2. Every output with a long name.
- 3. Only files ending in .clk.
- 4. Clock objects defined in the timing environment.

Week 08 | Question 45

What is the purpose of report_checks or an equivalent timing-report command?

- 1. Automatically submit a course assignment.
- 2. Change the chip's required behavior.
- 3. Display details of selected timing checks and paths.
- 4. Manufacture the chosen cell library.

Week 08 | Question 46

Why is reading a Liberty library important before link or timing analysis?

- 1. It proves all exceptions are valid.
- 2. It supplies the referenced cells and their timing models.
- 3. It routes every signal.
- 4. It assigns every design input a clock waveform automatically.

Week 08 | Question 47

Consider wildcard object queries in constraints. A. They can match no objects. B. They can match unintended additional objects. C. Their matches should be reviewed. Which statements are correct?

- 1. A, B and C
- 2. B only
- 3. A and C only
- 4. A and B only

Week 08 | Question 48

After technology-mapping changes, consider verification. A. Required functional equivalence matters. B. Relevant timing and electrical checks matter. C. Source-code line count alone establishes correctness. Which statements are correct?

- 1. A and B only
- 2. A only
- 3. B and C only
- 4. A, B and C

Week 08 | Question 49

Why can an area-optimal cover differ from a timing-optimal mapping?

- 1. Area and timing are mathematically identical metrics.
- 2. Cell delays, loading and logic depth affect timing beyond raw logic count.
- 3. Mapping ignores all electrical data.
- 4. The smallest cell is always the fastest under every load.

Week 08 | Question 50

What is a sound closure criterion for a constrained timing flow?

- 1. Hide every negative slack using false paths.
- 2. Stop after a single positive path report.
- 3. Meet the intended checks across relevant scenarios with justified constraints and reviewed exceptions.
- 4. Ignore all I/O timing because internal paths pass.

Week 09 - Power analysis and scan-based test

50 practice MCQs | one point per question | no negative marking in this practice set

Week 09 | Question 01

In the capacitive switching model P=alpha C V squared f, consider fixed other quantities. A. Power scales linearly with switched C. B. Power scales with voltage squared. C. Power is independent of frequency. Which statements are correct?

- 1. A and B only
- 2. B only
- 3. A, B and C
- 4. A and C only

Week 09 | Question 02

With C=10 fF, V=1 V, f=1 GHz and alpha=0.2, what is switching power?

- 1. 2 milliwatts
- 2. 20 microwatts
- 3. 0.2 milliwatt
- 4. 2 microwatts

Week 09 | Question 03

At unchanged C, V and alpha, frequency doubles. What happens to modeled switching power?

- 1. It doubles.
- 2. It halves.
- 3. It remains constant.
- 4. It becomes four times larger.

Week 09 | Question 04

At unchanged C, f and alpha, voltage falls from 1 V to 0.9 V. What is the switching-power ratio?

- 1. 0.19
- 2. 1.11
- 3. 0.90
- 4. 0.81

Week 09 | Question 05

At unchanged V, f and alpha, total switched capacitance falls by 25%. What is the power ratio?

- 1. 0.75
- 2. 0.25
- 3. 1.25
- 4. 1.75

Week 09 | Question 06

For a 2 mW block operating for 3 microseconds, what is the consumed energy if this is its average task power?

- 1. 0.67 nJ
- 2. 1.5 nJ
- 3. 6 nJ
- 4. 6 pJ

Week 09 | Question 07

Consider CMOS power components. A. Leakage can occur without logic transitions. B. Capacitive switching depends on activity. C. Leakage is necessarily zero whenever the clock is gated. Which statements are correct?

- 1. B and C only
- 2. A, B and C
- 3. A and B only
- 4. A only

Week 09 | Question 08

Consider short-circuit power during a CMOS transition. A. Pull-up and pull-down paths can conduct simultaneously for part of the transition. B. Input transition behavior can affect it. C. It is identical to the external capacitive charging term by definition. Which statements are correct?

- 1. A and B only
- 2. A and C only
- 3. B only
- 4. A, B and C

Week 09 | Question 09

Why can glitching increase dynamic power?

- 1. Extra transitions charge and discharge capacitances.
- 2. Every glitch necessarily changes the final steady-state truth table.
- 3. Glitching affects only printed waveform colors.
- 4. Glitches remove every switching event.

Week 09 | Question 10

What is clock gating intended to reduce?

- 1. Every leakage current by definition.
- 2. Unnecessary clock-related switching when a block need not update.
- 3. The number of manufactured wafers only.
- 4. The silicon substrate thickness.

Week 09 | Question 11

Why must clock gating be implemented carefully?

- 1. Any combinational AND gate guarantees glitch-free gating in all conditions.
- 2. An unsafe enable transition can create unwanted clock pulses.
- 3. Clock gating removes the need for reset.
- 4. Clock waveforms have no timing requirements.

Week 09 | Question 12

What does power gating target more directly than clock gating?

- 1. Only the number of input ports.
- 2. Leakage by disconnecting supply to an inactive block, with required isolation and state handling.
- 3. A mandatory increase in clock toggles.
- 4. Only source-code indentation.

Week 09 | Question 13

Why can lowering voltage reduce achievable performance?

- 1. The clock period is mathematically fixed by the file name.
- 2. Lower voltage always makes every cell faster.
- 3. Cell delays can increase as drive capability changes.
- 4. Voltage has no effect on transistor behavior.

Week 09 | Question 14

If task cycle count and voltage are fixed, halving frequency halves switching power and doubles runtime. What happens to modeled switching energy?

- 1. It becomes zero.
- 2. It stays unchanged in this simplified model.
- 3. It necessarily becomes four times larger.
- 4. It necessarily halves.

Week 09 | Question 15

What does a switching-activity trace provide?

- 1. A proof that all workloads have identical power.
- 2. Information about signal changes over a workload.
- 3. A list of wafer defects only.
- 4. A final GDS layout.

Week 09 | Question 16

Why does workload choice matter in power estimation?

- 1. Different input and state activity can produce different switching.
- 2. A waveform file changes the fabrication process.
- 3. Power is independent of every signal transition.
- 4. Every workload exercises all nets identically.

Week 09 | Question 17

For independent Boolean inputs with P(A=1)=0.5 and P(B=1)=0.4, what is P(A AND B=1)?

- 1. 0.60
- 2. 0.90
- 3. 0.10
- 4. 0.20

Week 09 | Question 18

For independent inputs with probabilities 0.5 and 0.4, what is P(A OR B=1)?

- 1. 0.30
- 2. 0.90
- 3. 0.20
- 4. 0.70

Week 09 | Question 19

For independent successive samples with P(signal=1)=0.5, what is the probability of a 0-to-1 transition?

1. 0.25
2. 0.75
3. 0.50
4. 1.00

Week 09 | Question 20

Under the same independent-sample model with P(signal=1)=0.2, what is the 0-to-1 transition probability?

1. 0.80
2. 0.20
3. 0.16
4. 0.04

Week 09 | Question 21

Why can a vectorless activity estimate be inaccurate?

1. Independence and activity assumptions may not match actual correlated signals.
2. It never uses signal probability assumptions.
3. It measures silicon current directly.
4. It always enumerates every real workload exactly.

Week 09 | Question 22

What does the internal-power model in a library account for beyond the external capacitive load model?

1. Only package mechanical stress.
2. Only the source file's memory size.
3. The retail price of the cell.
4. Energy consumed within the cell during specified events and conditions.

Week 09 | Question 23

Which technique can reduce unnecessary datapath switching?

1. Increasing all logic depths without reason.
2. Randomizing every idle input.
3. Operand isolation when the computation result is not needed.
4. Toggling every clock continuously.

Week 09 | Question 24

What is the objective of design for test (DFT)?

1. Eliminate every possible manufacturing defect.
2. Replace functional design verification entirely.
3. Prove all software correct.
4. Improve controllability and observability for manufacturing-fault testing.

Week 09 | Question 25

What is controllability in a test context?

- 1. How much power a package can deliver.
- 2. How readily a desired logic value can be established at a node.
- 3. How many source files a compiler reads.
- 4. How clearly a waveform color can be seen.

Week 09 | Question 26

What is observability in a test context?

- 1. The total number of masks.
- 2. The physical brightness of the chip surface.
- 3. How readily a node's effect can be observed at accessible outputs or scan elements.
- 4. The source-code line count.

Week 09 | Question 27

What does a scan flip-flop typically add compared with an ordinary flip-flop?

- 1. A guaranteed absence of clock-to-Q delay.
- 2. A controllable scan-data path and scan-enable selection.
- 3. An optical exposure source.
- 4. A full embedded operating system necessarily.

Week 09 | Question 28

In a conventional muxed-scan flip-flop, what is the usual role of scan enable?

- 1. Select shift data versus the functional data input.
- 2. Replace the reset signal's meaning in every design.
- 3. Adjust transistor doping dynamically.
- 4. Set the clock frequency to zero permanently.

Week 09 | Question 29

What typically happens during scan shifting?

- 1. The wafer is sliced into dies.
- 2. The entire circuit is verified against every specification.
- 3. Every flip-flop is removed from the netlist.
- 4. Serial test data moves through the scan chain.

Week 09 | Question 30

What typically happens during a functional capture phase of scan testing?

- 1. The circuit captures a response to the applied test state and inputs.
- 2. The package is disconnected from power permanently.
- 3. The cell library is rewritten.
- 4. All scan bits are forced to the same constant by definition.

Week 09 | Question 31

How many shift clocks load a complete 100-bit scan chain from a defined shift procedure?

- 1. 10
- 2. 1
- 3. 200
- 4. 100

Week 09 | Question 32

Two balanced 100-bit scan chains are loaded in parallel. How many shift clocks load both, ignoring capture overhead?

- 1. 200
- 2. 10,000
- 3. 50
- 4. 100

Week 09 | Question 33

What is a stuck-at-0 fault model?

- 1. The clock period is always zero.
- 2. A chip contains no registers.
- 3. A modeled signal remains at logic 0 regardless of its intended value.
- 4. A signal is always logic 1.

Week 09 | Question 34

What is a stuck-at-1 fault model?

- 1. A signal is always high impedance.
- 2. The power supply is physically always 0 V.
- 3. All outputs are removed from the design.
- 4. A modeled signal remains at logic 1 regardless of its intended value.

Week 09 | Question 35

What two steps are necessary to detect a single stuck-at fault through an output?

- 1. Only rename the fault site and rerun compilation.
- 2. Only add a comment in the RTL.
- 3. Activate the fault and propagate its effect to an observation point.
- 4. Only increase the clock period arbitrarily.

Week 09 | Question 36

For a two-input AND gate with output stuck at 0, which input detects the fault?

- 1. A=0, B=1
- 2. A=0, B=0
- 3. A=1, B=0
- 4. A=1, B=1

Week 09 | Question 37

For a two-input OR gate with output stuck at 1, which input detects the fault?

1. A=1, B=1
2. A=1, B=0
3. A=0, B=0
4. A=0, B=1

Week 09 | Question 38

For a fault on input A of an AND gate stuck at 0, what value of B allows propagation after A is activated?

1. B must be high impedance.
2. Either B value always works.
3. B=0
4. B=1

Week 09 | Question 39

What is the noncontrolling input value of an OR gate for fault propagation?

1. x
2. 0
3. 1
4. z

Week 09 | Question 40

For a 3-input NAND gate, what is the minimum test-set size to detect all single stuck-at faults on its inputs and output in the simple gate model?

1. 2
2. 8
3. 4
4. 3

Week 09 | Question 41

For a 5-input NAND gate under the same simple all-input/output single stuck-at model, the minimum test-set size is what?

1. 5
2. 32
3. 6
4. 10

Week 09 | Question 42

Why are random vectors sometimes inefficient for a rare-condition fault?

1. A random generator cannot produce binary values.
2. Random vectors remove the need for observation points.
3. The required activation and propagation combination can have low probability.
4. Every random vector is guaranteed to detect every fault.

Week 09 | Question 43

What does fault coverage measure?

- 1. The number of clock cycles per arithmetic operation.
- 2. The silicon area occupied by power rails.
- 3. The fraction of the stated target fault set detected, using the specified denominator.
- 4. The fraction of comments exercised.

Week 09 | Question 44

If 900 of 1000 listed target faults are detected and that full list is the denominator, what is fault coverage?

- 1. 10%
- 2. 111.1%
- 3. 90%
- 4. 99%

Week 09 | Question 45

Why should untestable faults be identified separately?

- 1. They affect interpretation of coverage and may not be detectable under the allowed model.
- 2. They must always be counted as detected without explanation.
- 3. Every untestable fault is automatically a fabricated failure.
- 4. They prove the complete design is functionally incorrect.

Week 09 | Question 46

What is a test point used for?

- 1. Guarantee zero extra area and no timing effect.
- 2. Improve access to control or observe difficult internal behavior.
- 3. Replace every scan register.
- 4. Only change the external package label.

Week 09 | Question 47

Consider scan insertion. A. The data-input selection structure can affect functional delay. B. Added loading can affect physical timing. C. Functional timing need not be rechecked because scan logic is logically separate. Which statements are correct?

- 1. A and B only
- 2. B and C only
- 3. A only
- 4. A, B and C

Week 09 | Question 48

Consider scan-shift and functional modes. A. Their clock requirements may differ. B. Scan-enable intent matters. C. One unconstrained mode automatically establishes the other mode's timing. Which statements are correct?

- 1. B only
- 2. A and C only
- 3. A, B and C
- 4. A and B only

Week 09 | Question 49

Which statement correctly distinguishes test coverage from functional correctness?

- 1. High manufacturing fault coverage does not prove the design implements the intended specification.
- 2. Functional verification makes every manufactured chip defect-free.
- 3. Detecting all listed faults proves every product requirement.
- 4. The two measures are identical in all flows.

Week 09 | Question 50

Which tool used in the course can estimate power using libraries and activity assumptions or traces?

- 1. OpenSTA.
- 2. A PDF viewer alone.
- 3. A standard text diff tool alone.
- 4. A silicon dicing saw.

Week 10 - ATPG, BIST and physical-design foundations

50 practice MCQs | one point per question | no negative marking in this practice set

Week 10 | Question 01

Consider ATPG. A. A target fault model is needed. B. Detection needs an observable response difference. C. ATPG necessarily proves the original product specification is correct. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. B and C only
- 4. A only

Week 10 | Question 02

Consider five-valued fault reasoning. A. D means good 1, faulty 0. B. Complemented D means good 0, faulty 1. C. D always denotes high impedance. Which statements are correct?

- 1. A and C only
- 2. A and B only
- 3. A, B and C
- 4. B only

Week 10 | Question 03

Consider fault activation. A. A stuck-at-0 site must be 1 in the good circuit. B. A stuck-at-1 site must be 0 in the good circuit. C. Activation alone guarantees observation through every downstream gate. Which statements are correct?

- 1. A, B and C
- 2. B and C only
- 3. A only
- 4. A and B only

Week 10 | Question 04

To activate a stuck-at-1 fault on node n, what should n be in the good circuit?

- 1. Any value without restriction
- 2. 0
- 3. 1
- 4. z necessarily

Week 10 | Question 05

To propagate an effect through a two-input AND gate, what should the other input be in the two-state model?

- 1. 0
- 2. z
- 3. An uninitialized x necessarily
- 4. 1

Week 10 | Question 06

To propagate an effect through a two-input OR gate, what should the other input be?

- 1. 1
- 2. An uninitialized x necessarily
- 3. z
- 4. 0

Week 10 | Question 07

A two-input NAND output is stuck at 1. Which input detects it?

- 1. 00
- 2. 11
- 3. 10
- 4. 01

Week 10 | Question 08

A two-input NOR output is stuck at 0. Which input detects it?

- 1. 01
- 2. 00
- 3. 11
- 4. 10

Week 10 | Question 09

What can block a fault effect at an AND gate output?

- 1. A noncontrolling 1 on the other input.
- 2. A correct observation point.
- 3. A controlling 0 on another input.
- 4. A fault activation at the target node.

Week 10 | Question 10

What is fault equivalence in test generation?

- 1. Every fault has the same physical cause.
- 2. Faults produce indistinguishable faulty behavior under the model and allowed observations.
- 3. Fault sites have identical printed names.
- 4. Faults occur on the same wafer necessarily.

Week 10 | Question 11

Why can fault collapsing reduce ATPG effort?

- 1. It physically repairs every defect.
- 2. It guarantees every random pattern detects every fault.
- 3. It removes all circuit inputs.
- 4. Equivalent or otherwise appropriately related faults can be represented by a reduced target set.

Week 10 | Question 12

What is a test cube?

- 1. A routing grid with no logical values.
- 2. A mandatory three-dimensional package shape.
- 3. A truth table containing only ones.
- 4. A partially specified test pattern with some don't-care input positions.

Week 10 | Question 13

Why may two compatible test cubes be merged?

- 1. Every don't-care must be converted to a fixed one first.
- 2. They must target the same source-code line.
- 3. All tests are compatible by definition.
- 4. Their specified input values do not conflict, allowing a combined test.

Week 10 | Question 14

What does backtracking in ATPG help do?

- 1. Increase the wafer diameter.
- 2. Route all clocks geometrically.
- 3. Automatically change the fault model to hide failure.
- 4. Revise decisions when the current assignments cannot achieve activation or propagation.

Week 10 | Question 15

What is built-in self-test (BIST)?

- 1. On-chip mechanisms for generating tests and evaluating responses.
- 2. A proof of every possible product behavior.
- 3. A rule that forbids external test equipment.
- 4. A package assembly technique only.

Week 10 | Question 16

Which pair is typical in logic BIST?

- 1. A pattern generator and a response compactor.
- 2. A source-code editor and a wafer saw only.
- 3. A photomask and a package lid only.
- 4. A clock constraint and a shipping invoice only.

Week 10 | Question 17

What is a linear feedback shift register (LFSR)?

- 1. A gate that performs optical exposure.
- 2. A memory whose size grows dynamically without bounds.
- 3. A shift register with linear feedback over binary arithmetic.
- 4. A substitute for every technology library.

Week 10 | Question 18

What is the maximum nonzero cycle length of an n-bit XOR-feedback LFSR with suitable primitive feedback?

- 1. n.
- 2. 2 to the power n minus 1.
- 3. 2 to the power n plus 1.
- 4. 2n.

Week 10 | Question 19

What is the maximum nonzero period of a 4-bit maximal-length LFSR?

- 1. 16
- 2. 4
- 3. 8
- 4. 15

Week 10 | Question 20

Why is an all-zero seed unsuitable for a conventional XOR-feedback LFSR?

- 1. It can remain locked in the all-zero state.
- 2. It always gives the maximum nonzero period.
- 3. It converts the shift register into an analog oscillator.
- 4. It doubles the number of storage bits.

Week 10 | Question 21

A 3-bit LFSR state is [s2,s1,s0] and next state is [s1,s0,s2 XOR s0]. What follows state 101?

- 1. 101
- 2. 001
- 3. 111
- 4. 010

Week 10 | Question 22

Under that same explicit recurrence, what follows state 001?

- 1. 000
- 2. 100
- 3. 101
- 4. 011

Week 10 | Question 23

What does a multiple-input signature register (MISR) do?

- 1. Measure final metal widths optically.
- 2. Guarantee that every faulty response has a unique signature.
- 3. Compact response streams from multiple inputs into a signature.
- 4. Replace all scan capture cycles with wafer probing.

Week 10 | Question 24

What is signature aliasing?

- 1. Different response sequences produce the same compacted signature.
- 2. Every response has a different signature.
- 3. A package has two identical printed labels.
- 4. The LFSR always stays in its seed state.

Week 10 | Question 25

Under an ideal uniformly random m-bit alias model, what is the approximate collision probability?

- 1. m divided by 2.
- 2. 1 divided by 2 to the power m.
- 3. Always zero.
- 4. 1 divided by m.

Week 10 | Question 26

Under that ideal model, a 16-bit signature has what approximate random-alias probability?

- 1. 1/32
- 2. 1/65,536
- 3. 1/256
- 4. 1/16

Week 10 | Question 27

If a fault is detected by one independent random trial with probability 0.25, what is the probability of detection within two trials?

- 1. 0.25
- 2. 0.0625
- 3. 0.4375
- 4. 0.50

Week 10 | Question 28

What is the probability that uniformly independent inputs of a 10-input AND are all 1?

- 1. 1/20
- 2. 1/10
- 3. 1/512
- 4. 1/1024

Week 10 | Question 29

Which fabrication task is ion implantation commonly used for?

- 1. Introducing controlled dopants into semiconductor regions.
- 2. Selecting a Verilog sensitivity list.
- 3. Reordering a scan chain in a timing graph.
- 4. Compressing a response into a signature.

Week 10 | Question 30

What is associated with front-end-of-line fabrication?

- 1. Only the final system software update.
- 2. Only drawing package labels.
- 3. Only source-code parsing.
- 4. Forming transistor/device structures.

Week 10 | Question 31

What is associated with back-end-of-line fabrication?

- 1. Only generating random ATPG seeds.
- 2. Only compiling C into an executable.
- 3. Building interconnect metal and dielectric layers above devices.
- 4. Only the initial product requirements meeting.

Week 10 | Question 32

What is the role of an interlayer dielectric?

- 1. Electrically separate conductors except at intended connections.
- 2. Store the final RTL text permanently.
- 3. Short every crossing wire by design.
- 4. Replace all transistors.

Week 10 | Question 33

Under R=R_sheet times L/W, what is wire resistance for R_sheet=0.1 ohm per square, L=200 micrometres and W=2 micrometres?

- 1. 10 ohms
- 2. 0.001 ohm
- 3. 100 ohms
- 4. 40 ohms

Week 10 | Question 34

Under the same sheet-resistance model, doubling width with length fixed has what effect?

- 1. Resistance stays identical.
- 2. Resistance becomes four times larger.
- 3. Resistance halves.
- 4. Resistance doubles.

Week 10 | Question 35

For a uniform wire with unchanged cross-section, both R and C scale with length. Doubling length changes the simple RC product by what factor?

- 1. 2
- 2. 4
- 3. 1/2
- 4. 8

Week 10 | Question 36

What causes capacitive crosstalk between neighboring nets?

- 1. Electrical coupling through mutual capacitance.
- 2. Identical source-code indentation.
- 3. A shared package marketing name.
- 4. The choice of PDF viewer.

Week 10 | Question 37

Why can the relative switching direction of an aggressor matter?

- 1. Crosstalk exists only in ideal zero-capacitance wires.
- 2. It can change the effective coupling impact on the victim's delay or noise.
- 3. All aggressors affect a victim identically regardless of transitions.
- 4. Opposite transitions remove every physical field.

Week 10 | Question 38

What is the manufacturing antenna effect concerned with?

- 1. The number of source files in an RTL hierarchy.
- 2. Charge accumulation on conductors damaging gate oxide during processing.
- 3. A normal requirement that every clock radiate a radio signal.
- 4. Only the shape of the external package.

Week 10 | Question 39

Which measure can help mitigate a process antenna problem, depending on rules and flow?

- 1. Renaming the gate.
- 2. Replacing every input vector by zero.
- 3. Antenna diodes or an appropriate routing-layer jump.
- 4. Deleting all verification constraints.

Week 10 | Question 40

Which physical data format commonly describes routing layers, rules and cell abstracts?

- 1. VCD.
- 2. LEF.
- 3. A plain SDC clock report only.
- 4. A C testbench executable.

Week 10 | Question 41

Where is routing-layer sheet resistance commonly provided in a technology abstract used by a physical tool?

- 1. Only an RTL module declaration.
- 2. Only the package's retail label.
- 3. Technology LEF layer data.
- 4. Only a waveform signal list.

Week 10 | Question 42

Which statement separates LEF and Liberty?

- 1. Both are always the same file with different extensions.
- 2. Liberty contains only final top-level metal polygons.
- 3. LEF supplies physical abstract information; Liberty supplies characterized cell logic and electrical models.
- 4. LEF contains no physical information.

Week 10 | Question 43

What does the RC of an interconnect affect?

- 1. Signal delay, transition behavior and sometimes noise interactions.
- 2. The total number of possible Boolean functions independently of implementation.
- 3. Only the spelling of net names.
- 4. Only how many comments are in RTL.

Week 10 | Question 44

Two identical independent vias of resistance R are placed in ideal parallel. Their equivalent resistance is what?

- 1. R squared
- 2. R/2
- 3. 2R
- 4. R

Week 10 | Question 45

What is electromigration associated with?

- 1. A change in the Boolean meaning of AND.
- 2. The migration of RTL files between folders.
- 3. Material transport driven by high current density over time.
- 4. The number of random simulation seeds only.

Week 10 | Question 46

For fixed current, doubling conductor cross-sectional area changes current density by what factor?

- 1. 1
- 2. 4
- 3. 1/2
- 4. 2

Week 10 | Question 47

Consider local decoupling capacitance. A. It can supply transient charge. B. It can help reduce supply disturbance within the network's limits. C. It eliminates all supply resistance. Which statements are correct?

- 1. A and B only
- 2. A and C only
- 3. A, B and C
- 4. B only

Week 10 | Question 48

Consider well or substrate taps. A. They provide required bias connections. B. Their use can depend on process rules. C. They are a replacement for the functional Boolean specification. Which statements are correct?

- 1. B and C only
- 2. A only
- 3. A and B only
- 4. A, B and C

Week 10 | Question 49

What are filler cells commonly used for in a standard-cell layout?

- 1. Guarantee minimum placement wire length.
- 2. Implement arbitrary new functional RTL behavior.
- 3. Fill permitted gaps and maintain required row-level physical continuity.
- 4. Replace every timing buffer.

Week 10 | Question 50

Which tool in the course supports an RTL-to-GDS physical implementation flow?

1. OpenROAD.
2. A Unix file-copy command alone.
3. A general PDF renderer.
4. GTKWave alone.

Week 11 - Chip planning, power delivery and placement

50 practice MCQs | one point per question | no negative marking in this practice set

Week 11 | Question 01

Consider floorplanning. A. Macro arrangement matters. B. Routing and power space matter. C. Functional equivalence alone determines every physical coordinate. Which statements are correct?

- 1. A and C only
- 2. B only
- 3. A, B and C
- 4. A and B only

Week 11 | Question 02

Consider a standard-cell core region. A. It is a physical implementation region inside the die. B. Its usable area can be reduced by macros or blockages. C. It is simply the sum of RTL source-file sizes. Which statements are correct?

- 1. B and C only
- 2. A and B only
- 3. A only
- 4. A, B and C

Week 11 | Question 03

Consider core-to-die planning margins. A. Boundary and I/O needs can matter. B. Power structures can need space. C. Every process and design requires exactly the same margin. Which statements are correct?

- 1. A and B only
- 2. B only
- 3. A, B and C
- 4. A and C only

Week 11 | Question 04

If standard-cell area is 60,000 square micrometres and the available placement region is 100,000 square micrometres, utilization is what?

- 1. 40%
- 2. 6%
- 3. 60%
- 4. 166.7%

Week 11 | Question 05

For 80,000 square micrometres of cells at a target 50% utilization in an otherwise unobstructed region, required region area is what?

- 1. 80,000 square micrometres
- 2. 200,000 square micrometres
- 3. 160,000 square micrometres
- 4. 40,000 square micrometres

Week 11 | Question 06

Why is targeting 100% standard-cell utilization often impractical?

- 1. Routing, placement flexibility and other physical needs require space.
- 2. Cells do not occupy physical area.
- 3. Full utilization always guarantees minimum delay.
- 4. Every library forbids any placement above 10%.

Week 11 | Question 07

What is aspect ratio in a rectangular floorplan when defined as height/width?

- 1. The number of cells divided by die cost.
- 2. Clock period divided by input delay.
- 3. The core height divided by its width.
- 4. Total wire resistance divided by voltage.

Week 11 | Question 08

A rectangular core has area 40,000 square micrometres and height/width=1. What are its dimensions?

- 1. 400 by 400 micrometres.
- 2. 20 by 20 micrometres.
- 3. 200 by 200 micrometres.
- 4. 100 by 400 micrometres.

Week 11 | Question 09

A core is 100 by 200 micrometres. What is height/width if height is 200?

- 1. 100
- 2. 2
- 3. 20,000
- 4. 0.5

Week 11 | Question 10

Why consider macro pin locations during macro placement?

- 1. Pins have no relationship to connected nets.
- 2. Macro rotation never affects access.
- 3. Every macro pin is automatically shorted to every other pin.
- 4. Pin access and connectivity influence routing feasibility and delay.

Week 11 | Question 11

What is a placement halo around a macro commonly used for?

- 1. Record its source file's date.
- 2. Define an extra scan state bit.
- 3. Add a new Boolean output.
- 4. Reserve separation or keepout space for access and physical constraints.

Week 11 | Question 12

What is a macro channel?

- 1. Space between large blocks that can provide routing or other required resources.
- 2. A manufacturing wafer defect.
- 3. A Verilog always block.
- 4. A software message queue necessarily.

Week 11 | Question 13

Why can abutting macros without access planning be problematic?

- 1. Connected pins and routing resources may become inaccessible or congested.
- 2. Macro edges automatically form every required signal connection.
- 3. Physical space never affects routing.
- 4. Pin shapes are unrelated to connectivity.

Week 11 | Question 14

What is a placement blockage?

- 1. A physical region restricting placement of selected objects.
- 2. A source-code syntax error only.
- 3. A required all-zero scan pattern.
- 4. A property that proves no timing path exists.

Week 11 | Question 15

What is a routing blockage?

- 1. A guarantee that every net is already connected.
- 2. A restriction on routing resources in a specified region or layer.
- 3. A prohibition on all logic simulation.
- 4. An instruction to delete the netlist.

Week 11 | Question 16

What is a primary purpose of a power delivery network (PDN)?

- 1. Replace the signal-routing network entirely.
- 2. Define every state-transition relation.
- 3. Deliver power and ground to cells with acceptable electrical behavior.
- 4. Execute all testbench instructions.

Week 11 | Question 17

Why are mesh or grid PDN structures useful?

- 1. They remove every need for power analysis.
- 2. Multiple current paths can improve distribution robustness and lower effective delivery impedance.
- 3. They make wire resistance exactly zero.
- 4. They guarantee no routing-resource trade-off.

Week 11 | Question 18

What is static IR drop?

- 1. Voltage drop caused by current through resistive supply paths.
- 2. A mandatory clock phase difference.
- 3. The number of registers in a binary counter.
- 4. A measure of source-code indentation.

Week 11 | Question 19

A supply path has resistance 0.2 ohm and current 0.1 A. What is its IR drop?

- 1. 2 mV
- 2. 200 mV
- 3. 20 mV
- 4. 2 V

Week 11 | Question 20

With a 1.0 V source and a 40 mV resistive drop, what voltage reaches the modeled load?

- 1. 1.04 V
- 2. 0.96 V
- 3. 0.60 V
- 4. 0.04 V

Week 11 | Question 21

Which change reduces resistive drop for fixed current in a simple isolated stripe model?

- 1. Increase stripe cross-section or provide suitable parallel paths.
- 2. Rename the power net.
- 3. Increase resistance.
- 4. Narrow every stripe without changing anything else.

Week 11 | Question 22

Why can a denser PDN affect signal routing?

- 1. Power conductors occupy routing resources.
- 2. PDN wires exist outside all physical layers.
- 3. Signal and power networks never compete for geometric space.
- 4. Routing ignores power structures entirely.

Week 11 | Question 23

What is global placement primarily trying to optimize?

- 1. Only the chip's packaging invoice.
- 2. Only final via manufacturing.
- 3. The source-code line ending format.
- 4. A coarse cell arrangement considering objectives such as wire length, density and timing.

Week 11 | Question 24

Can a global-placement solution temporarily contain cell overlap?

- 1. Yes, legalization may be required afterward.
- 2. Overlap is a Boolean truth-table property.
- 3. Only if every cell is a macro.
- 4. No, every global method directly produces a final signoff layout.

Week 11 | Question 25

What is legalization?

- 1. Convert every design to a full-custom transistor schematic.
- 2. Generate a scan test signature.
- 3. Change the product requirements.
- 4. Remove placement overlap and meet applicable row/site and placement restrictions.

Week 11 | Question 26

What is detailed placement used for?

- 1. Write the initial C algorithm.
- 2. Replace all technology rules with random choices.
- 3. Refine legal cell locations and improve physical objectives locally.
- 4. Fabricate the silicon ingot.

Week 11 | Question 27

What is half-perimeter wire length (HPWL) for one net?

- 1. The sum of the rectangle's four sides.
- 2. The squared Euclidean distance of every pin pair.
- 3. The exact routed length in every case.
- 4. The x-span plus y-span of the pin bounding box.

Week 11 | Question 28

A net's pins span x=10 to 70 and y=20 to 50 micrometres. What is its HPWL?

- 1. 30 micrometres
- 2. 90 micrometres
- 3. 60 micrometres
- 4. 180 micrometres

Week 11 | Question 29

Why is HPWL only an estimate of routed wire length?

- 1. It explicitly routes every wire and via.
- 2. It is computed only after detailed routing.
- 3. It includes all possible coupling capacitance exactly.
- 4. It omits actual routing topology, detours, layers and obstacles.

Week 11 | Question 30

What can a high cell-density hotspot cause?

- 1. Guaranteed elimination of timing violations.
- 2. Guaranteed lower wire length for every net.
- 3. Congestion and reduced placement/routing flexibility.
- 4. Zero need for legalization.

Week 11 | Question 31

What is routability-driven placement concerned with?

- 1. Only choosing testbench seeds.
- 2. Whether the arrangement leaves enough usable routing resources for the nets.
- 3. Only printing source code.
- 4. Only the alphabetical ordering of instances.

Week 11 | Question 32

What is timing-driven placement concerned with?

- 1. Only changing net names.
- 2. Only the total wafer count.
- 3. Improving path timing through physical arrangement and related optimization.
- 4. Only minimizing the number of PDF pages.

Week 11 | Question 33

Why can placing connected cells closer help a critical net?

- 1. It always eliminates every coupling interaction.
- 2. It may reduce wire length and parasitic delay, subject to other constraints.
- 3. It always removes cell delay.
- 4. It changes the required Boolean output automatically.

Week 11 | Question 34

What does scan-chain reordering commonly optimize physically?

- 1. The algorithm's arithmetic precision.
- 2. The number of manufacturing mask layers.
- 3. The connection order of scan cells to reduce shift-chain wiring cost.
- 4. The intended functional combinational truth table.

Week 11 | Question 35

Why must scan-chain reordering preserve the test flow's interpretation?

- 1. Scan patterns do not depend on chain ordering.
- 2. Reordering automatically removes all scan clocks.
- 3. Pattern generation and scan connections must agree on the final chain order.
- 4. Every order uses the same serialized bit positions without adjustment.

Week 11 | Question 36

What are spare cells placed for?

- 1. Replace all original functional cells immediately.
- 2. Execute user software as a CPU necessarily.
- 3. Facilitate later engineering changes without a complete new placement plan.
- 4. Guarantee zero die area.

Week 11 | Question 37

Why can spare-cell placement location matter?

- 1. A later ECO needs accessible nearby resources and routable connections.
- 2. Spare cells are not physical objects.
- 3. All future changes occur at the same coordinate.
- 4. Their connectivity is determined only by their file names.

Week 11 | Question 38

What is an engineering change order (ECO) in implementation?

- 1. A controlled modification to an existing design implementation.
- 2. A mandatory redesign of every transistor from scratch.
- 3. A new independent course assignment.
- 4. Only a manufacturing shipment label.

Week 11 | Question 39

What should follow an ECO affecting logic or routing?

- 1. Assume all old signoff evidence remains valid without review.
- 2. Recheck relevant functionality, timing and physical rules.
- 3. Ignore parasitic changes.
- 4. Delete all constraints.

Week 11 | Question 40

In hierarchical design, why define block interface timing budgets?

- 1. To make physical interconnect delay irrelevant.
- 2. To coordinate block-level closure with top-level timing requirements.
- 3. To eliminate top-level timing analysis.
- 4. To prove every block independently of all interfaces.

Week 11 | Question 41

Why can a block meeting its own timing budget still need top-level reanalysis?

- 1. A block has no interfaces after integration.
- 2. Every hierarchy boundary deletes delay.
- 3. Top-level analysis measures only source-code indentation.
- 4. Actual integration clocks, loads and interconnect may differ from assumed conditions.

Week 11 | Question 42

What information from LEF is directly useful for placement?

- 1. Cell dimensions, pin shapes and physical obstructions.
- 2. Only every software runtime result.
- 3. Only dynamic testbench output.
- 4. Only the target's sales forecast.

Week 11 | Question 43

What does a DEF file commonly describe in an implementation exchange flow?

- 1. Design-specific physical placement and routing information.
- 2. A waveform history only.
- 3. A high-level C algorithm only.
- 4. Only the unmodified process technology manual.

Week 11 | Question 44

Which report family helps assess aggregate timing violations in OpenROAD?

- 1. Reports such as report_tns and worst-slack reports.
- 2. Only a chip packaging report.
- 3. Only a histogram of source-code characters.
- 4. Only a directory listing.

Week 11 | Question 45

Three endpoint slacks are -0.02 ns, -0.03 ns and +0.04 ns. Summing the negative endpoints gives what TNS?

- 1. -0.05 ns
- 2. -0.01 ns
- 3. -0.09 ns
- 4. +0.05 ns

Week 11 | Question 46

Why re-evaluate congestion after adding timing buffers?

- 1. Routing depends only on the initial floorplan title.
- 2. Timing repair automatically guarantees routability.
- 3. Buffers have no area or pin demand.
- 4. New cells and nets consume physical and routing resources.

Week 11 | Question 47

Consider placement sites and rows. A. They define alignment and legal placement structure. B. Legalization respects applicable site/row restrictions. C. Geometric overlap is acceptable at final signoff whenever logic is equivalent. Which statements are correct?

- 1. A only
- 2. B and C only
- 3. A, B and C
- 4. A and B only

Week 11 | Question 48

Consider endcap and well-tap planning. A. Physical requirements can consume space. B. Required placement can be process-dependent. C. These cells make every signal route unnecessary. Which statements are correct?

- 1. A, B and C
- 2. A and C only
- 3. A and B only
- 4. B only

Week 11 | Question 49

What can happen when a large macro is rotated?

- 1. Its documented Boolean functionality must always invert.
- 2. Its logical input count necessarily doubles.
- 3. Its silicon process automatically changes.
- 4. Its physical pin locations and access conditions change with the orientation.

Week 11 | Question 50

Which is a sound floorplan trade-off?

- 1. Treat power structures as having no geometric footprint.
- 2. Balance area, utilization, macro access, power delivery, timing and routability.
- 3. Minimize all whitespace regardless of process needs.
- 4. Maximize density while ignoring all routing and timing.

Week 12 - Clock trees, routing, extraction and signoff

50 practice MCQs | one point per question | no negative marking in this practice set

Week 12 | Question 01

Consider clock timing. A. Insertion delay needs a defined source and sink. B. Equal insertion delays at two sinks imply zero pairwise skew under the same reference. C. Every sink's insertion delay equals the data-path cycle count. Which statements are correct?

- 1. A and B only
- 2. B and C only
- 3. A only
- 4. A, B and C

Week 12 | Question 02

Define skew as capture arrival minus launch arrival. Clock arrivals are 0.55 ns at capture and 0.35 ns at launch. What is skew?

- 1. +0.20 ns
- 2. +0.90 ns
- 3. +0.35 ns
- 4. -0.20 ns

Week 12 | Question 03

Two sinks each have clock insertion delay 1 ns relative to the same reference. What is their pairwise skew?

- 1. 2 ns
- 2. 0 ns
- 3. 1 ns
- 4. -1 ns

Week 12 | Question 04

Consider physical clock buffering. A. Sink capacitance loads the clock network. B. Wires affect clock delay and slew. C. Correct logical toggling proves ideal physical delivery. Which statements are correct?

- 1. A and B only
- 2. A, B and C
- 3. B only
- 4. A and C only

Week 12 | Question 05

Consider CTS objectives. A. Skew and insertion delay can matter. B. Slew, loading and power can matter. C. A CTS pass alone proves the product algorithm correct. Which statements are correct?

- 1. B and C only
- 2. A, B and C
- 3. A only
- 4. A and B only

Week 12 | Question 06

Why may a geometrically symmetric H-tree still have nonzero electrical skew?

- 1. H-trees cannot contain buffers.
- 2. Loads, parasitics, cell variation and routing details can differ.
- 3. Geometric symmetry guarantees identical electrical conditions in every real design.
- 4. Electrical skew depends only on the number of RTL comments.

Week 12 | Question 07

What is one trade-off of a clock mesh compared with a simple tree?

- 1. Potentially robust clock delivery at the cost of substantial wiring and switching load.
- 2. Guaranteed absence of any routing demand.
- 3. Complete removal of clock drivers.
- 4. Guaranteed zero clock power.

Week 12 | Question 08

What does useful skew attempt to exploit?

- 1. Randomly change every clock period.
- 2. Replace every data path with a clock wire.
- 3. Remove all hold checks from signoff.
- 4. Intentional relative clock timing to improve selected timing margins with other checks preserved.

Week 12 | Question 09

With capture-minus-launch skew convention, positive useful skew can have what trade-off?

- 1. Better setup margin but reduced hold margin.
- 2. Worse setup and better hold margin always.
- 3. No timing effect at all.
- 4. Better setup and hold margin by the same amount always.

Week 12 | Question 10

Nominal setup slack is 60 ps at zero skew. Capture arrives 30 ps later relative to launch, with other quantities fixed. New setup slack is what?

- 1. 30 ps
- 2. -90 ps
- 3. 90 ps
- 4. -30 ps

Week 12 | Question 11

Nominal hold slack is 30 ps at zero skew. Capture arrives 30 ps later relative to launch, with other quantities fixed. New hold slack is what?

- 1. 30 ps
- 2. 60 ps
- 3. -60 ps
- 4. 0 ps

Week 12 | Question 12

Setup slack at zero skew is 60 ps and hold slack is 30 ps. With k=capture-minus-launch and other values fixed, which skew interval satisfies both ideal checks?

- 1. -30 ps to +60 ps.
- 2. Any positive skew without a bound.
- 3. +60 ps to +30 ps.
- 4. -60 ps to +30 ps.

Week 12 | Question 13

Why should clock gating cells and their enables receive timing checks?

- 1. Clock-gating timing applies only to source-code compilation.
- 2. The gating operation must avoid unsafe clock pulses and respect control timing.
- 3. Gated clocks can never glitch under any enable transition.
- 4. Gating eliminates the need for clock distribution.

Week 12 | Question 14

What is a clock-tree sink?

- 1. A wafer defect map.
- 2. A source-code file's last line.
- 3. Every signal port regardless of use.
- 4. A destination clock pin or specified endpoint of the clock network.

Week 12 | Question 15

Source latency is 2.0 ns and modeled network latency is 0.6 ns. What is their total under an additive convention?

- 1. 2.6 ns
- 2. 1.4 ns
- 3. 0.3 ns
- 4. 3.2 ns

Week 12 | Question 16

What does global routing primarily produce?

- 1. The original untimed algorithm.
- 2. A manufacturing test response signature.
- 3. A coarse routing plan allocating paths and resources.
- 4. Every final legal wire polygon and via in all cases.

Week 12 | Question 17

What does detailed routing add beyond global routing?

- 1. Only an ideal zero-resistance connectivity graph.
- 2. Only a package marketing plan.
- 3. Exact tracks, layers, wire segments and vias subject to detailed rules.
- 4. Only a list of function names.

Week 12 | Question 18

When does routing overflow occur on a resource?

- 1. Demand equals zero.
- 2. Capacity exceeds demand.
- 3. The netlist has more than one module.
- 4. Demand exceeds its available capacity.

Week 12 | Question 19

An edge has capacity 8 tracks and demand 11 tracks. What is its overflow?

- 1. 8 tracks
- 2. 19 tracks
- 3. 3 tracks
- 4. 11 tracks

Week 12 | Question 20

Two routing edges have (demand,capacity)=(7,5) and (6,8). Sum positive overflow on the edges.

- 1. 2
- 2. 0
- 3. 4
- 4. 26

Week 12 | Question 21

What does zero global-routing overflow establish by itself?

- 1. Every detailed design rule necessarily passes.
- 2. All paths necessarily meet timing.
- 3. The coarse capacity model has no overflow, but detailed legality still needs checking.
- 4. No vias are needed in the design.

Week 12 | Question 22

Why can a global route require a detour?

- 1. Routing layers have infinite capacity.
- 2. Connectivity is independent of endpoints.
- 3. Obstacles or congested resources can prevent the shortest coarse path.
- 4. Every routed net must be a straight Euclidean line.

Week 12 | Question 23

For rectilinear routing with no obstacles, what is the Manhattan distance from (3,5) to (10,1)?

- 1. 11 units
- 2. 7 units
- 3. 4 units
- 4. About 8.06 units

Week 12 | Question 24

What is the role of a via in multilayer routing?

- 1. Join every nearby crossing net.
- 2. Connect conductors across selected metal layers.
- 3. Set every sink's data value to zero.
- 4. Replace the need for a timing library.

Week 12 | Question 25

Why can redundant vias improve reliability?

- 1. They remove every need for physical rules.
- 2. They change an AND function into an OR function.
- 3. They guarantee no routing area cost.
- 4. Multiple valid connections can reduce dependence on one via and may lower resistance.

Week 12 | Question 26

What must redundant-via insertion respect?

- 1. Applicable geometry, spacing and connectivity constraints.
- 2. Only source-code line length.
- 3. No process-dependent rules.
- 4. Only a clock period label.

Week 12 | Question 27

What does a routing DRC violation indicate?

- 1. The Boolean function is necessarily wrong on every input.
- 2. A geometric implementation condition fails a specified design rule.
- 3. The source-code file cannot be parsed.
- 4. Every clock has zero skew.

Week 12 | Question 28

Why are minimum spacing rules needed?

- 1. To meet process-dependent manufacturability and electrical separation requirements.
- 2. To guarantee equal delays on all nets.
- 3. To create unknown values in simulation.
- 4. To force all connected pins apart logically.

Week 12 | Question 29

What is a routing rip-up-and-reroute step?

- 1. Remove selected route segments and try an alternative implementation.
- 2. Recompile the operating system only.
- 3. Dismiss every DRC without fixing it.
- 4. Erase the intended circuit behavior.

Week 12 | Question 30

Why can routing-order choices affect congestion?

- 1. Routes never share physical regions.
- 2. Every net has its own unlimited private routing layer.
- 3. Earlier nets can occupy resources needed by later nets.
- 4. Net order determines the truth table directly.

Week 12 | Question 31

What does parasitic extraction derive?

- 1. The minimum Boolean cover from a clock waveform alone.
- 2. The original software source from a package label.
- 3. Electrical resistance and capacitance associated with physical interconnect.
- 4. The manufacturing exam score.

Week 12 | Question 32

What is a common use of SPEF in implementation analysis?

- 1. Define the complete RTL algorithm.
- 2. Exchange extracted parasitic network information.
- 3. Record only user interface colors.
- 4. Store a processor executable.

Week 12 | Question 33

Why is post-route extraction useful for timing?

- 1. Actual wire geometry can yield more realistic RC than early estimates.
- 2. It guarantees zero uncertainty.
- 3. It replaces all functional verification.
- 4. It removes every cell timing arc.

Week 12 | Question 34

Under a simple uniform-wire model, R scales with length. If length increases 50% with cross-section unchanged, resistance scales by what?

- 1. 2.25
- 2. 0.5
- 3. 1/1.5
- 4. 1.5

Week 12 | Question 35

A driver sees 6 fF of pin load and 9 fF of wire capacitance. Ignoring coupling, total load is what?

- 1. 15 fF
- 2. 54 fF
- 3. 9 fF
- 4. 3 fF

Week 12 | Question 36

For a two-section RC tree with ground capacitances C1 at the first node and C2 at the sink, which Elmore expression applies at the sink, ignoring driver resistance?

- 1. R1(C1+C2)+R2C2.
- 2. (R1+R2)/(C1+C2).
- 3. R1C1+R2C1.
- 4. R1C2 only.

Week 12 | Question 37

Using that two-section model, R1=2 kilohms, R2=1 kilohm, C1=3 fF and C2=5 fF. What is the Elmore time constant?

- 1. 15 ps
- 2. 21 ps
- 3. 11 ps
- 4. 30 ps

Week 12 | Question 38

Why can coupling capacitance require more context than a fixed ground capacitance?

- 1. Its effect can depend on neighboring-net transitions.
- 2. It is determined only by RTL variable names.
- 3. It always cancels wire resistance completely.
- 4. It has no possible effect on delay or noise.

Week 12 | Question 39

What does LVS verify after routing?

- 1. Extracted layout connectivity matches the intended reference within the check's scope.
- 2. Every path meets setup and hold.
- 3. All silicon defects are detected.
- 4. The design uses minimum power.

Week 12 | Question 40

Why does a DRC-clean result not establish timing closure?

- 1. DRC always computes every clock-domain property.
- 2. Timing depends only on geometry-rule count.
- 3. Geometry rules and timing requirements are different checks.
- 4. A legal wire has exactly zero delay.

Week 12 | Question 41

What does tapeout release for manufacture?

- 1. An arbitrary unconstrained intermediate netlist.
- 2. Only a course certificate.
- 3. Final approved design data, commonly including a GDSII or equivalent layout database.
- 4. Only the testbench's console output.

Week 12 | Question 42

What is a GDSII stream primarily used to represent?

- 1. Only a register-transfer algorithm.
- 2. A complete software runtime trace.
- 3. Only timing exceptions.
- 4. Geometric layout data with hierarchy and layers.

Week 12 | Question 43

Why should final signoff use the final routed design version?

- 1. Version identity affects only file naming.
- 2. Old analysis always proves every future change.
- 3. Every route change is electrically identical by definition.
- 4. Late ECOs and route changes can invalidate earlier analysis results.

Week 12 | Question 44

A post-route hold path has slack -12 ps. A proposed data buffer adds at least 20 ps delay without other changes. What is its new ideal hold slack?

- 1. 32 ps
- 2. -32 ps
- 3. -8 ps
- 4. 8 ps

Week 12 | Question 45

A setup path has slack 18 ps. Its reroute adds 25 ps maximum delay. All else fixed, new setup slack is what?

- 1. -7 ps
- 2. 43 ps
- 3. 7 ps
- 4. -43 ps

Week 12 | Question 46

Why should setup and hold both be rechecked after a clock-tree ECO?

- 1. Clock changes affect only source-code compilation.
- 2. Clock sinks are independent of timing constraints.
- 3. A setup pass necessarily proves hold.
- 4. Relative launch/capture clock changes can affect the checks in opposite directions.

Week 12 | Question 47

Consider a late buffer ECO. A. Added cells need legal physical locations. B. New connections can affect DRC and timing. C. Previous signoff automatically remains valid without reanalysis. Which statements are correct?

- 1. A and B only
- 2. B only
- 3. A, B and C
- 4. A and C only

Week 12 | Question 48

Consider antenna cleanup. A. Current routing must meet applicable process antenna rules. B. A layer jump or diode may be a valid remedy in the appropriate flow. C. Timing closure alone proves all antenna checks pass. Which statements are correct?

- 1. A only
- 2. A, B and C
- 3. A and B only
- 4. B and C only

Week 12 | Question 49

What is the purpose of maintaining final constraint and parasitic provenance?

- 1. Replace all numerical analysis with file names.
- 2. Guarantee every source file was written by one person.
- 3. Ensure signoff results correspond to the intended clocks, exceptions and physical network.
- 4. Remove the need for design review.

Week 12 | Question 50

Which statement best describes signoff readiness?

- 1. No extracted interconnect is ever needed after routing.
- 2. Required functional, timing, electrical and physical checks pass for the intended design version and scenarios.
- 3. A clean source-code compile guarantees manufacture-ready layout.
- 4. One favorable setup report is sufficient for every requirement.